Approach for Optimization of Smart Network-Integrated V2G Control Strategies

Thesis in partial fulfillment of the requirements for the degree
Master of Science (M.Sc.)

in the course of studies
M.Sc. Elektrotechnik

submitted by
Laurenz Bareiß

Matriculation number
407280

Under the scientific supervision of
Prof. Dr. Dr. h.c. Sahin Albayrak

Under the scientific guidance of
Dominik Ascher, M.Sc.

Berlin, April 2024

Technische Universität Berlin
Fakultät IV – Elektrotechnik und Informatik
DAI-Labor
Fachgebiet AOT

Affidavit

- ✓ Die Arbeit enthält eine deutsche Zusammenfassung.
- ✓ Die Arbeit enthält eine englische Zusammenfassung.
- ✓ Die Arbeit enthält eine Selbständigkeitserklärung nach § 60 Abs. 8 AllgStuPO:

Hiermit erkläre ich an Eides statt, dass ich die vorliegende Arbeit selbstständig und eigenhändig sowie ausschließlich unter Verwendung der aufgeführten Quellen und Hilfsmittel angefertigt habe.

Laurenz Bareiß
Berlin, 14. April 2024

- ✓ The thesis contains a German abstract.
- ✓ The thesis contains an English abstract.
- ✓ The thesis includes a declaration that I completed the work independently in accordance with section § 60 Abs. 8 AllgStuPO:

I hereby confirm that I prepared this thesis independently and by exclusive reliance on literature or tools indicated herein.

Laurenz Bareiß
Berlin, April 14, 2024

Abstract

Deutsch

Mit der zunehmenden Elektrifizierung des Mobilitätssektors und der digitalen Transformation des Energienetzes und -marktes entstehen an der Schnittstelle dieser Sektoren neue technische Möglichkeiten, die gegenseitige Vorteile versprechen. Eine dieser Möglichkeiten ist der Einsatz von Vehicle-to-Grid (V2G). In dieser Arbeit wird ein umfassender Optimierungsansatz für V2G Kontrollstrategien untersucht, der in eine realistische und netzwerkintegrierte Smart Grid Umgebung eingebettet ist. Mit dem Aufkommen von Smart Homes, Smart Metering und Smart Grid-Technologien sind Endverbraucher zu aktiven Teilnehmern des Energiemarktes und der Verteilnetze geworden. Diese Entwicklung begann mit der zunehmenden Integration von Photovoltaikanlagen und hat sich im Laufe der Zeit auf intelligentes Lastmanagement und den Einsatz von Energiespeichersystemen im privaten Kontext zur Erhöhung des Eigenverbrauchs ausgeweitet. Neuerdings wird auch die aktive Teilnahme der Endverbraucher am Energiemarkt durch die Möglichkeit der Direktvermarktung ihrer Erzeugung oder sogar der Teilnahme am Regelenergiemarkt immer beliebter. Die zunehmende Möglichkeit, den persönlichen Energiebedarf zu steuern, hat auch zur Verbreitung flexibler Energietarife auf der Basis von Day-Ahead-Marktpreisen geführt. Der nächste Baustein auf dem Weg zu intelligenten und nachhaltigen Energienetzen ist die Integration zunehmender Batteriekapazitäten, die größtenteils ungenutzt auf der Straße abgestellt werden. Die Nutzung von V2G hat in anderen Ländern bereits Einzug erhalten, ist aber im europäischen Kontext eher Forschungsthema. Das Ziel dieser Thesis ist die Entwicklung eines umfassenden Ansatzes zur Optimierung von V2G-Steuerungsstrategien auf Basis aktueller und zukünftiger technologischer Trends in Energiesystemen. Das Hauptziel besteht darin durch die Anwenung von

Reinforcement-Learning, Potentiale für Verbraucher auszuschöpfen, die Netzstabilität zu erhöhen, die Energienutzung zu optimieren und somit eine nachhaltige Energienutzung zu fördern. Durch die Einbettung des Ansatzes in realistische Szenarien, die auf realen Ausgangsdaten basieren, wird die potenzielle zukünftige Bedeutung der V2G-Integration aufgezeigt. Die Ergebnisse unterstreichen die Bedeutung der Entwicklung adaptiver, intelligenter Kontrollmechanismen, um das volle Potenzial der V2G-Technologie auszuschöpfen und den Weg für eine effizientere, nachhaltigere und verbraucherfreundlichere Energiezukunft zu ebnen.

Schlüsselwörter: *V2G-Optimierungsstrategien; Smart-Grid-Integration; Nachhaltige Energielösungen*

English

With the increasing electrification of the mobility sector and the digital transformation of the energy grid and market, new technical opportunities are emerging at the intersection of these evolving sectors, promising mutual benefits. One such opportunity is the deployment of vehicle-to-grid (V2G). It serves both the customer who owns the electric vehicle and the grid that uses it for grid services. This thesis explores a comprehensive approach to optimize control strategies embedded in a realistic and integrated smart grid environment. With the emergence of smart homes, advanced metering, and smart grid technologies, end users have become active contributors and participants in the energy market and distribution networks. This transition began with the increasing integration of distributed photovoltaic (PV) systems and has evolved over time towards smart load management and the use of energy storage systems to increase self-consumption. More recently, the active participation of end users in the energy market through the ability to directly market their production or even participate in the electricity balancing market has gained popularity. The increasing ability to manage personal energy demand has also led to the popularity of flexible energy tariffs based on day-ahead market prices. The next evolving keystone in the transition to smart and sustainable energy grids is the integration of the growing amount of battery capacity parked on the street. In other countries, the use of V2G has already arrived, while in the European context it is more of a research topic. The aim of this study is to develop a comprehensive

approach to optimize V2G control strategies based on these recent and future technological trends within the energy system domain. By applying Reinforcement Learning, the main objectives of this work are to create customer benefits, grid stability, optimize energy utilization and thus promote sustainable energy practices. By embedding the simulation in realistic scenarios based on real world data, the potential future role of V2G integration is demonstrated. The results highlight the importance of developing adaptive, intelligent control mechanisms to realize the full potential of V2G technology and pave the way for a more efficient, sustainable and consumer empowered energy future.

Keywords: *V2G Optimization Strategies; Smart Grid Integration; Sustainable Energy Solutions*

Contents

List of Figures i

List of Tables ii

List of Algorithms iii

List of Abbreviations v

1 Introduction 1
- 1.1 Motivation . . . 1
- 1.2 Problem Statement . . . 3
- 1.3 Objectives . . . 4
- 1.4 Research Questions . . . 5
- 1.5 Contributions . . . 6
- 1.6 Outline of Work . . . 7

2 Theoretical Foundations 9
- 2.1 Reinforcement Learning Foundation . . . 9
  - 2.1.1 Finite Markov Decision Processes . . . 9
  - 2.1.2 Fundamental Concepts of Reinforcement Learning . . . 11
  - 2.1.3 Reinforcement Learning Algorithms . . . 14
    - 2.1.3.1 Q-Learning . . . 14
    - 2.1.3.2 Deep Deterministic Policy Gradient . . . 15
- 2.2 Power Grid and Power Flow Fundamentals . . . 17
  - 2.2.1 Structure and Components of Power Grids . . . 17
  - 2.2.2 Power Flow Calculation . . . 19

3 Methodology 22
- 3.1 Research Design and Conceptual Framework . . . 22
  - 3.1.1 Overview . . . 23
- 3.2 Training Data Collection . . . 26
  - 3.2.1 Energy Price: Day-Ahead-Market . . . 27
  - 3.2.2 PV Profiles . . . 29
  - 3.2.3 Load Profiles . . . 31
  - 3.2.4 Initial Charging Station and ESS Values . . . 32

3.3 Conceptual Model and Assumptions 35
3.3.1 Model Architecture 36
3.3.2 Assumptions 38

4 Implementation 39
4.1 Software / Platform Selection and Rationale 39
4.2 Software Design and Implementation Details 41
4.2.1 Integration with Power Network Calculations 46
4.2.2 Tracking of Partial Penalties and Rewards 46

5 OptCharge Reinforcement Learning Environment 48
5.1 Network Operation Logic 48
5.2 State, Action, and Reward Definitions 49
5.2.1 States 50
5.2.2 Actions 55
5.2.3 Penalties and Rewards 57
5.2.4 Balancing of Reward-Signal for Distinct Optimization Goals 62
5.3 Environment Overview 63
5.3.1 Initialization of the Environment 64
5.3.2 Reset-Function 66
5.3.3 Step-Function 67

6 Testing and Validation of Approach 69
6.1 Scope and metrics for evaluation 69
6.1.1 Evaluation Setting 69
6.1.2 Evaluation Metrics 70
6.2 Scenarios with Isolated Rewards 73
6.3 Testing Evaluation Summary 85

7 Demonstration Developed Approach 86
7.1 Scenario Selection 86
7.1.1 Scenario Design and Case Studies 87
7.1.2 Evaluation of Scenario Results 90

8 Discussion and Conclusion 92
8.1 Recommendations for Future Research 94

A User Manual 95

B Experimental Penalty 96

List of References 99

Index **103**

List of Figures

2.1 Agent–Environment Interaction in a MDP [6, p. 46] 11

3.1 Schematic Overview of an Example Network 24
3.2 Distribution Day Ahead 28
3.3 Solar Irradiance and Resulting PV Production 31
3.4 Example Load Profiles of 4 Random Profiles and Days 32
3.6 Conceptual Overview of the Framework 36

6.1 CS-Schedule | Isolated Penalty p_{pv} and p_{grid} 75
6.2 Training statistics and Results | Penalties: PV and Grid-Demand . 76
6.3 CS-Schedule | Isolated Penalty p_{cs}, p_{pv} and p_{grid} 77
6.4 Training statistics and Results | Penalties: PV, Grid-Demand and SoC 79
6.5 V2G-Schedule | Isolated Penalty p_{cs}, p_{grid} and Reward r_{grid} | (c) considers ESS 81
6.6 V2G Training statistics and Results | Penalties: Grid-Demand, SoC and Reward: Grid | ESS included in (c) 82
6.7 Line penalty p_{line} Training statistics and Results 84

7.1 Training Stats Scenario Low 2035, High 2035 and Utopia 2035 . . 90

List of Tables

3.1 Network Configuration Options 25
3.3 Combined Arrival and Departure Time Probabilities 33

5.1 Environment Actions 55

6.1 DDPG-Configuration 73
6.2 Network Configuration PV 75
6.3 Network Configuration V2G 80

7.1 NEP: Scenarios 87
7.2 NEP-Scenario: Low 2035 Translated to a 10-Bus Network 88
7.3 NEP-Scenario: High 2035 Translated to a 10-Bus Network 89
7.4 Scenario: Utopia 2035 Translated to a 10-Bus Network 89

List of Algorithms

3.1 Determine Presence and SoC on arrival for Cahrging Stations . . . 34

5.1 Applied Power Calculation $P_{\mathrm{cs,t}}$ for a CS from action $a_{\mathrm{cs},t}$ 56

List of Abbreviations

CPO	Charge Point Operator
CS	Charging Station
DA	Day-Ahead
DDPG	Deep Deterministic Policy Gradient
EMSP	E-Mobility Service Provider
EPEX	European Power Exchange
ESS	Energy Storage System
EV	Electric Vehicle
GTI	Global Tilted Irradiance
MDP	Markov Decision Process
NEP	German Network Development Plan
PV	Photovoltaic
RL	Reinforcement Learning
SB3	Stable Baselines3
SoC	State of Charge
STC	Standard Test Conditions
V2G	Vehicle-to-Grid
V2H	Vehicle-to-Home

1 Introduction

The transition towards more sustainable and efficient energy systems presents both challenges and opportunities, particularly in the integration of *Electric Vehicles* (EVs) into the power grid. The increasing adoption of EVs, coupled with advances in *Vehicle-to-Grid*s (V2Gs) applications, offers the opportunity not only to improve grid stability and optimize energy consumption, but also to further the goals of decarbonizing the energy sector through optimized use of renewable generation. This chapter lays the groundwork for exploring an *Approach for Optimization of Smart Network-Integrated V2G Control Strategies*. It begins with the Motivation Section 1.1, emphasizing the importance of embedding EVs in energy systems beyond their use in transportation. Subsequent sections address the current challenges and potential solutions associated with V2G integration, as detailed in Section 1.2, outline the primary objectives in Section 1.3, continue with the derived research questions in Section 1.4, and discuss the contributions in Section 1.5. Through this investigation, this thesis aims to contribute to the evolving discourse on sustainable energy systems by providing insights and strategies to harness the potential of EVs within smart operating energy grids.

1.1 Motivation

As a measure of decarbonization and efficiency within the transportation sector, the number of EVs continues to grow, with Germany alone aiming to have around 15 million EVs on the road by 2030.[1] Interestingly, studies estimate that 90% of daily trips for private cars are less than 40 km, and on average, cars in general spend more than 23 hours a day parked.[2] This makes it reasonable to look for other meaningful ways to use EVs for purposes other than just driving. Since EV batteries still account for about one-third of vehicle acquisition costs [3], it is

worth looking for alternative, profitable, and even grid-supportive ways to use them. When connected to a *Charging Station* (CS), an EV can also potentially be used as a *Energy Storage System* (ESS), provided they are technically capable of doing so. Either for *Vehicle-to-Home* (V2H) purposes, or by integrating EVs into the grid, bidirectional charging, i.e., charging and discharging the battery on demand in form of V2G, could be an important building block in the transformation of the energy grid. This has been highlighted, for example, in the EU Strategy for Energy System Integration [4] and the German government's Integrated National Energy and Climate Plan [5]. In light of these factors, the combination of EVs and their charging infrastructure emerges as an ideal flexible resource through which V2G can support energy management, demand response and grid stability, while optimizing the use of renewable energy sources and minimizing costs. Potential use cases for V2G include optimization of local (self-)consumption, grid-friendly charging to support distribution grid load, marketing of energy and provision of other system services (e.g. ancillary services/peak shaving). This can create added value for *Charge Point Operator*s (CPOs), *E-Mobility Service Provider*s (EMSPs) and customers, through financial benefits from a temporary increase in self-supply as well as from variable tariffs, feed-in payments or compensation for the provision of ancillary services. In addition, the high power and energy capacity of car batteries compared to average stationary ESS enables them to respond quickly with high power and large amounts of energy to changes in the grid.

Although V2G is not yet widespread in Germany and Europe, mainly due to a past lack of interest on the part of local and locally relevant car manufacturers, manufacturers are increasingly starting to integrate V2G functionality. In Germany, unidirectional smart charging has been tested in various projects, both on a small scale for individual households and in more centralized approaches. As the EV infrastructure expands, it offers potential for further V2G capabilities. While current V2G studies and trials focus primarily on unidirectional smart charging to mitigate load peaks through staggered charging or reduced load for longer periods, more advanced approaches are being explored.

The resulting customer opportunities and general economic opportunities are summarized in the following

Customer Opportunities:

- Monetary benefits from (temporary) increase in self-supply or marketing of energy at the point of connection to the grid.
- Monetary benefits from variable tariffs and marketing of flexibility.
- Enabling "energy-in-quarters" concepts or other "extended self-supply" norms.
- Increased security of supply (islanding capabilities).

Economic Opportunities:

- **Local** - Volatile energy prices, grid fees, levies.
  Customers can monetarily optimize their interface with the energy supplier by optimizing consumption. Additional options arise when combined with self-generation (e.g. solar panels) or stationary storage.
- **Regional** - Avoid grid expansion.
  Secure, controllable charging power and the aggregation of multiple charging points can optimize distribution grid expansion and reduce the need for low-voltage grid expansion.
- **Supra-regional** - Grid services, energy trading.
  Flexibility in the power grid is becoming increasingly important with the expansion of renewable energy. EV-batteries can provide regulation power. Transmission system operators gain access to a large pool of energy storage options, reducing system service costs and grid charges for customers.

1.2 Problem Statement

Residential load management, including EVs, has recently gained popularity and has become a focus of scientific research for its integration and optimization in this domain. A key interest is to increase self-consumption, primarily by charging the EV, for example, during periods of high power generation from *Photovoltaic* (PV)- and renewable-sources. Although the use of smart charging is gradually increasing in line with variable electricity tariffs, it is mainly used for unidirectional charging, which involves the regulation and scheduling of EV charging based on user behavior. The ongoing digitization of the electricity grid, facilitated by

connected metering systems and emerging regulatory and technical standards, is also paving the way for more sophisticated V2G-standards and charging strategies. In the future, smart charging may not only address the goal of increasing self-consumption or purchasing and storing electricity at low cost, but may also include the ability to return energy to the grid. This could be further monetized by selling the energy during peak electricity price periods as part of a collective or even as an individual contributor, allowing for arbitrage trading.

There is a clear need to explore new methodologies for EV charging strategies, especially considering V2G and ancillary services potential. As the technology is still in the research phase in Europe, except for a few pilot projects, real application of V2G remains largely unexplored. As progress continues in terms of cheaper hardware, improved batteries, standardization and regulatory changes, V2G may soon become a critical part of the transition to a more renewable energy supply. However, there is potential in providing decision support and methodologies that holistically address the consumer, economic, and network opportunities associated with bidirectional charging. Numerous projects are currently underway to optimize V2G charging schedules using a variety of strategies. Whether the strategy is classical (non-)linear programming, optimal control, or reinforcement learning, the impact of these plans on the physical power system often remains unexplored. Although some efforts have been made, there is considerable room for further exploration of methodologies that effectively combine optimization strategies with validation of network effects. In this context, this thesis attempts to delve deeper into these complexities. Its primary goal is to identify potential opportunities and develop a decision support framework that enables the creation of optimized charging and discharging schedules for individual electric vehicles and fleets. A key focus will be to develop a holistic and realistic method for bidirectional charging that effectively balances optimization strategies with a high-level examination of their impact on the network.

1.3 Objectives

The overall goal of this thesis is to explore and improve the integration of EV into the distribution grid considering V2G control strategies. This includes the develop-

ment of a comprehensive decision support framework that enables the creation of optimized charging and discharging schedules for individual EVs. A major focus will be on developing a comprehensive approach that embeds the bidirectional charging capabilities of V2G in realistic scenarios of small-scale distribution networks. The strategy should not only maximize the efficiency of energy use, but also assess the impact of such strategies on the power balance of the electric grid, taking into account the dynamic interplay between energy supply and demand. The research aims to investigate the potential economic, grid, and customer benefits for various stakeholders in the EV and energy sectors, including EV owners, grid operators, and the broader energy ecosystem. By conducting in-depth simulations and analyses, the thesis aims to validate the proposed optimization strategies, ensuring their practical applicability and effectiveness in real-world scenarios. Through this exploration, the thesis aims to contribute meaningful insights and solutions that support the sustainable integration of EVs into the energy grid, fostering a more efficient and resilient energy landscape.

1.4 Research Questions

From the above problem statement and objectives, the research questions that will guide the development of the approach are as summarized as follows: How can a comprehensive, intelligent charging optimization methodology for electric vehicle charging points be specified, developed and implemented? How can this method balance a variety of optimization goals, respect network constraints through appropriate depth of network simulation, offer potential for V2G services, and contribute to customer, economic, and network opportunities? Finally, what are the potential benefits of this method for stakeholders in the electric vehicle and energy grid ecosystem? Based on this, the research will delve into more detailed questions, the answers to which will be explored and substantiated through the research process, with subsequent contributions emerging later in the study:

1) How to specify a comprehensive smart charging optimization problem for EV charging points that considers various optimization objectives and serves as a decision aid for charging and discharging processes? How to cover V2G use cases and include grid simulation for validation? Which solution approaches

can be considered for the given problem? What tools and methods should be used to support the development and implementation?

2) How can an approach be developed from the previous problem specification to balance different optimization goals, e.g. to ensure that all EVs are sufficiently charged by the end of the day, to minimize peak load, to increase renewable self-consumption, to consider congestion management, to provide V2G services, and to minimize electricity costs, while respecting network constraints, by validating the developed charging plans through network simulation of appropriate depth? How can this be embedded in sufficiently realistic scenarios, taking into account typical load profiles of producers and consumers?

3) How can sufficiently realistic scenarios be represented, modelled and solved, and what consumer, economic and network opportunities can be derived from this approach? What are the potential benefits of the proposed V2G optimization approach for customers and the overall electric vehicle and energy network ecosystem?

1.5 Contributions

Derived from the three-part classification of the aforementioned research questions in Section 1.4, the following contributions result, which constitute the goal of this thesis. The core contribution of this research is the development of an "Approach for Optimization of Smart Network-Integrated V2G Control Strategies". In order to constructively build on this, the research contributions will be carried out and structured as follows:

1) **Problem Specification**
Development of a comprehensive optimization problem specification - preferably modular (allowing different strategies/techniques) - that allows the optimization of (dis-)charging processes in v2g usage for a variety of EVs. The resulting decision support tool should achieve optimal V2G scheduling while prioritizing different objectives and taking into account the network underlying the scenario. Criteria for the selection of appropriate tools

and methodologies to support the development and implementation of this methodology will be identified.

2) **Development Approach / Framework**
Development of a comprehensive framework/methodology resulting from 1) that balances multiple optimization goals for electric vehicle charging, e.g. ensuring adequate daily charging, increasing renewable self-consumption, minimizing peak load, managing congestion, providing V2G services, reducing electricity costs through arbitrage trading, while improving grid utilization/respecting grid constraints. The methodology to be developed aims at validating the optimized charging plans through network simulations in realistically approximated scenarios.

3) **Demonstration of the developed approach / framework**
Demonstration of the developed approach/framework by creating sufficiently realistic scenarios to serve as input. Potentially, the possible use of different techniques to solve the formulated problem should allow to compare them in their performance.

1.6 Outline of Work

After the purpose of this work was clarified in Chapter 1, Chapter 2 lays the theoretical foundation for a comprehensive model. First, the basic principles of RL are established, followed by an overview of the necessary calculations for power flow and line loads in distribution networks.

Once the basics are established, by going through the reseachs Methodology a conceptual model is developed in Chapter 3. This phase addresses methodological issues, structural considerations, and data analysis to ensure that optimization is embedded in an appropriate realistic design framework.

Moving from conceptualization to implementation, the discussion shifts to how to effectively segment and execute the model, leading to the identification of appropriate software modules.

The implementation phase is dedicated to the computation of essential variables for modeling and the detailed structuring of the environment, including the definition

of states, actions, observations, and rewards, along with the internal logic for their computation.

Upon completion, the model's functionality is validated through simple examples and a comparison of different algorithms is performed.

Based on these analyses, the approach is applied to small but realistic scenarios of future technology development stages for evaluation purposes.

The narrative concludes with a speculative look at potential advances in research in this area, culminating in a conclusion.

2 Theoretical Foundations

This section introduces the basic principles required for the developed approach. *Reinforcement Learning* (RL)-optimization will be used to optimize within a distribution grid environment, the necessary basics and principles of the RL-domain are first introduced in Section 2.1, and then the necessary background for the calculations in distribution grids is introduced in Section 2.2.

2.1 Reinforcement Learning Foundation

The concept and application of RL is widespread within optimization problems that are difficult to formulate in classical optimization approaches such as linear programming. Since the formulated problem of optimizing v2g charging schedules is dependent on various states of its environment, in this thesis a well established RL method is used to solve the formulated problem. In this section, Introduction to RL, the fundamental concept of RL is briefly introduced.

Beginning with an explanation of the underlying theory of RL in *Markov Decision Processes* (MDPs), the more detailed mechanisms of RL will be discussed in order to finally break it down into problem-specific terms and also to present the algorithms used for decision support in an abstract form.

2.1.1 Finite Markov Decision Processes

This section briefly describes the problem of finite MDPs. MDPs serve as a fundamental concept in the domain of RL. Finite MDPs model the sequential nature of decision-making, where each choice affects not only immediate outcomes, but also future states and their associated rewards. MDPs encapsulate the challenge of

balancing immediate rewards against future benefits, a critical aspect of strategic decision-making. *Sutton*, in his 'Reinforcement Learning Introduction' [6], gives a comprehensive overview of the basic concepts within the RL domain.

Finite MDPs extend the idea of evaluative feedback by incorporating the associative process of selecting actions in different contexts. Unlike simpler models that focus on determining the value of an action in isolation, MDPs require assessing the value of actions within particular states and the value of the states themselves under optimal action choices. This involves calculating two key metrics: the value of taking a particular action a in a particular state s, denoted by $q_*(s, a)$, and the value of being in a state s, denoted by $v_*(s)$, assuming optimal action choices. These calculations are essential for understanding the long-term effects of decisions and for accurately attributing outcomes to specific actions.[6] At every discrete moment, the system has a certain state s, whereupon the decision maker can select an action a accessible from state s. In response, the system transitions to a new state s' at the subsequent time step, awarding the decision maker a specific reward $R_a(s, s')$.

The likelihood of the system transitioning to state s' is influenced by the action taken, as described by the transition probability function $P_a(s, s')$. Therefore, the future state s' is determined by the present state s and the action a taken by the decision maker. Once s and a are known, the future state is independent of past states and actions, demonstrating the Markov property that underpins state transitions in a MDP.

MDPs expand upon Markov chains by incorporating actions, which introduce decision-making capabilities, and rewards, which provide incentives for certain actions. In contrast, if there is only a single possible action for each state, such as "wait", and all rewards are uniform, such as "zero", then the MDP simplifies to a Markov chain.

MDPs are a mathematically idealized form of the reinforcement learning problem. Key concepts such as returns, value functions, and Bellman equations, which offer a structured approach to analyzing and resolving MDPs are subsequently introduced. These concepts are integral to grasping how present decisions can impact future rewards and navigating the complexities of sequential decision-making.

2.1.2 Fundamental Concepts of Reinforcement Learning

The explained MDPs serve as a frame for the challenge of learning through interaction towards achieving a goal. The entity that learns and functions as a decision maker is termed the *agent*. Everything external to the agent, with which it interacts, is called the *environment*. The following is a description of these interactions.

The Agent–Environment Interface

The agent and the environment engage in a continuous loop of interactions: the agent chooses actions, and the environment responds to those actions by presenting new circumstances to the agent. The environment is also responsible for providing rewards, which are scalar values. The agent's goal is to maximize these rewards over time by strategically choosing its actions. [6, p. 47] The process by which the agent interacts with the environment is depicted in Figure 2.1.

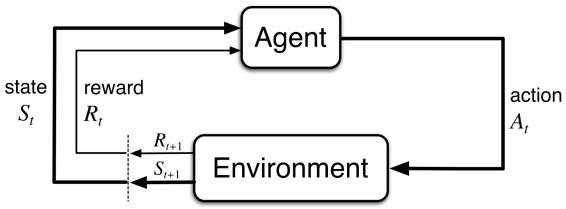


Fig. 2.1: Agent–Environment Interaction in a MDP [6, p. 46].

To break down the agent-environment interaction in a MDP from Figure 2.1 more clearly, interactions between the agent and the environment occur at discrete intervals in a series, denoted by $t = 0,1,2,3,\ldots$. In our case, only at each discrete interval t the agent is presented with a representation of the current *state* of the environment, denoted by $S_t \in \mathcal{S}$, on the basis of which it chooses an *action*, represented by $A_t \in \mathcal{A}(s)$. In the next time step, as a direct or indirect result of the chosen action, the agent is presented with a reward, $R_{t+1} \in \mathcal{R} \subset \mathbb{R}$, and the environment transitions to a new state, S_{t+1}, based on the actions taken. [6, p. 48] This interaction between the MDP and the agent generates a trajectory or sequence that begins as follows:

$$S_0, A_0, R_1, S_1, A_1, R_2, S_2, A_2, R_3, \ldots \tag{2.1}$$

In the context of a finite MDP, the collections of states, actions, and rewards ($\mathcal{S}, \mathcal{A}$, and $\mathcal{R}$) are each bounded to a finite set of elements. Under these conditions, the random variables R_t and S_t have well-defined distinct discrete probability distributions that depend only on the immediate prior state and action. Consequently, for given instances of these random variables, where $s' \in \mathcal{S}$ and $r \in \mathcal{R}$, there exists a defined probability that these specific values will manifest at time t, given certain prior state and action values. [6, p. 48]

$$p\left(s', r \mid s, a\right) \doteq \Pr\left\{S_t = s', R_t = r \mid S_{t-1} = s, A_{t-1} = a\right\} \tag{2.2}$$

Whereas any $s', s \in \mathcal{S}$, $r \in \mathcal{R}$, and $a \in \mathcal{A}(s)$. The p function defines the dynamics of the MDP. The equation uses an overdot on the equality sign to indicate that it is a definition. Function $p : \mathcal{S} \times \mathcal{R} \times \mathcal{S} \times \mathcal{A} \rightarrow [0,1]$, which maps its four arguments to the interval [0,1], is an ordinary deterministic function. The '|' symbol is associated with conditional probability; however, its use here is to emphasize that p specifies a probability distribution corresponding to each pair of s and a [6, p. 49], which means that:

$$\sum_{s' \in \mathcal{S}} \sum_{r \in \mathcal{R}} p\left(s', r \mid s, a\right) = 1, \text{ for all } s \in \mathcal{S}, a \in \mathcal{A}(s). \tag{2.3}$$

In a Markov decision process, the transition dynamics of the environment are fully encapsulated by the probabilities denoted by p. Specifically, the probability of landing in a given state S_t and receiving a given reward R_t is influenced only by the immediately preceding state S_{t-1} and action A_{t-1}. Importantly, this probability is independent of any previous states or actions. This concept primarily imposes a condition on the definition of the state rather than on the decision process itself. For a state to be effective within this framework, it must encapsulate all relevant information from past interactions between the agent and the environment that could potentially affect future outcomes. A state that satisfies this criterion possesses the Markov property.

Using the dynamics function p, which has four arguments, it is possible to derive any additional information about the environment that may be needed. This includes the state transition probabilities, which are represented in a slightly unconventional

way as a function p with three parameters. This function maps combinations of states and actions, denoted as $\mathcal{S} \times \mathcal{A} \times \mathcal{S}$, to a probability range between 0 and 1, expressed as $p : \mathcal{S} \times \mathcal{A} \times \mathcal{S} \rightarrow [0,1]$. [6, p. 50]

$$p\left(s' \mid s, a\right) \doteq \Pr\left\{S_t = s' \mid S_{t-1} = s, A_{t-1} = a\right\} = \sum_{r \in \mathcal{R}} p\left(s', r \mid s, a\right) \tag{2.4}$$

It is possible to determine the expected rewards associated with particular state-action pairings by using a function r that takes two arguments. This function is defined as $r : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$, where it maps each state-action pair to a value of expected reward.

$$r(s, a) \doteq \mathbb{E}\left[R_t \mid S_{t-1} = s, A_{t-1} = a\right] = \sum_{r \in \mathcal{R}} r \sum_{s' \in \mathcal{S}} p\left(s', r \mid s, a\right) \tag{2.5}$$

In addition, a three-parameter function r is used to compute the expected rewards corresponding to combinations of a state, an action, and the subsequent state. This function, written as $r : \mathcal{S} \times \mathcal{A} \times \mathcal{S} \rightarrow \mathbb{R}$, assigns each state-action-next-state triplet a real number representing the expected reward.

$$r\left(s, a, s'\right) \doteq \mathbb{E}\left[R_t \mid S_{t-1} = s, A_{t-1} = a, S_t = s'\right] = \sum_{r \in \mathcal{R}} r \frac{p\left(s', r \mid s, a\right)}{p\left(s' \mid s, a\right)} \tag{2.6}$$

The framework of Markov Decision Processes (MDPs) abstracts to a significant degree the challenge of learning goal-directed behaviors through interaction. It posits that, regardless of the specifics involved in sensing, remembering, and controlling, and regardless of the goals pursued, any learning task aimed at acquiring goal-directed actions can essentially be distilled into the exchange of three types of signals between an agent and its environment: a signal for the agent's decisions (the actions), a signal that forms the basis for those decisions (the states), and a signal that outlines the agent's goals (the rewards). While this framework may not effectively capture every nuance of decision-making and learning scenarios, its broad applicability and utility have been demonstrated in various contexts. [7]

2.1.3 Reinforcement Learning Algorithms

Based on the knowledge gained about the agent-environment interface, it is still necessary to determine how the agent optimizes its decision strategy. For this purpose, the agent is provided with algorithms appropriate for the problem to be solved. In the approach developed here, an algorithm is required for application to a continuous action space. The algorithm used to optimize the developed environment is *Deep Deterministic Policy Gradient* (DDPG), which can be thought of as deep Q-learning for continuous action spaces. In the following, Q-learning is briefly explained before the algorithm of DDPG is abstractly described.

2.1.3.1 Q-Learning

The explained Q-Learning theory corresponds to [8] which the implementation of the used optimization framework *Stable Baselines3* (SB3) is based on.

In Q-learning, the agent learns a policy π that maps states to actions, aiming to maximize cumulative rewards. The core component of Q-learning is the Q-function, denoted as $Q(s, a)$, which estimates the expected utility of taking an action a in state s and following the optimal policy thereafter. The Q-function is updated iteratively using the Bellman equation as follows:

$$Q(s, a) \leftarrow Q(s, a) + \alpha[r + \gamma \max_{a'} Q(s', a') - Q(s, a)] \tag{2.7}$$

Whereas:

- s is the current state.
- a is the current action.
- r is the reward received after taking action a in state s.
- s' is the new state after action a is taken.
- a' is a possible future action taken in the new state s'.
- α is the learning rate, determining to what extent the newly acquired information will override the old information.

- γ is the discount factor, representing the difference in importance between future rewards and present rewards.

This equation updates the Q-value by blending the old value with the new information, which includes the reward and the discounted maximum future reward. The term $\max_{a'} Q(s', a')$ represents the best possible future reward obtainable from the new state s', and by continuously updating the Q-values in this way, the agent learns an optimal policy π^* that maximizes the expected cumulative reward.

2.1.3.2 Deep Deterministic Policy Gradient

The DDPG strategy extends the principles of Q-learning, specifically designed for environments characterized by continuous action spaces. Similar to Q-learning, DDPG seeks to optimize the agent's total reward in a given environment. However, it adapts the conventional Q-learning framework for continuous action domains by integrating deep learning techniques and policy gradient methods, thus facilitating the handling of complex, continuous action spaces. [9]

In essence, DDPG employs two neural networks: the critic and the actor. The critic network approximates the Q-function, evaluating the expected return from taking certain actions in specific states. Simultaneously, the actor network is tasked with determining the optimal action for a given state. This division into actor and critic components enables DDPG to efficiently manage the intricacies of continuous action spaces. [9]

The critic network refines its parameters by minimizing the discrepancy between the predicted Q-values and the target Q-values, which are derived using the Bellman equation, akin to traditional Q-learning methods. These target Q-values, a combination of immediate rewards and discounted future rewards, reflect the critic's estimation of the value of current state-action pairs.

Conversely, the actor network updates its parameters through a policy gradient approach, steering the parameters in a direction that maximizes the expected future rewards, as estimated by the critic. This optimization is achieved by computing the gradient of the critic's output with respect to the actor's parameters, and employing gradient ascent to enhance the actor's policy.

In summary, DDPG harnesses the predictive capabilities of Q-learning for future rewards, while utilizing a deterministic policy gradient method to navigate the continuous action space. Through iterative updates to the actor and critic networks, DDPG adeptly learns policies that maximize cumulative rewards. This dynamic interplay between exploration of the environment and exploitation of known information facilitates optimal decision-making in complex, continuous domains.

2.2 Power Grid and Power Flow Fundamentals

In order to implement a network-integrated optimization approach, some elementary basics about the used components have to be introduced. For this purpose, the basic principles for the required integration of the components in their interaction are first established, before they are briefly explained individually. The theoretical foundations that are then introduced and required for the calculations within the distribution network as part of this approach are limited to the basics of load flow calculation and the resulting line loading of each line in the network.

2.2.1 Structure and Components of Power Grids

In this section the basic structure of the power grid, outlining its key components is given. A representation of a distribution network, where the approach is embedded in, with their components and their dynamics is essential. The modeled grid is based on the standard 400 V low voltage level. Generally, a distribution network can be thought of as a collection of individual nodes, each of which can draw power from or feed power into the distribution grid. The nodes, resp. buses are representing individual points in the network that can act as power sources or sinks depending on power dynamics of the components connected to them. The connection of these buses via lines results in a representation of a distribution network.

In this model, only the active power of all components is considered, which simplifies the calculation in many places, but is sufficient in the context of the experience to be gained. Of the used components there is only the *External Grid* – connecting the distribution grid of our environment with the outer medium voltage grid – and *Static Generator* – representing the PV-components in the environment – where power values are based on the generator viewpoint (positive active power means power generation). For the remaining used components of bus, load and storage, the signing is based on the consumer viewpoint (positive active power means power consumption). [10]

External Grid

The external grid element represents a voltage source.[10] In this approach, it can be simplified acting as a power sink or source, depending on the overall power balance across the individual buses, to absorb excess power or supply power in the event of a deficit, thus ensuring a stable power balance within the simulated network. In theory, distribution networks can be connected to the external network at multiple nodes, but in the approach developed, only one external network is connected to a configurable bus.

Bus

Buses represent the nodes of the network, connecting individual technologies such as load, PV, CS and ESS components, and are therefore representative of individual households in the simulated distribution network.

In the context of the surrounding disribution grid, the bus combines all incoming and outgoing power flows of its components and balances them through its connection to other buses. The resulting bus power of bus i in timestep t is defined as $P_{\text{bus,i}}(t)$

Load

Loads are used to model power consumption. Here they are meant to represent typical household load profiles. They are reduced to their active power. The load power is defined by the simulated scenario in the current time step t at bus i as $P_{\text{load, i}}(t)$.

Static Generator

Static generators are used to model power injection with active power. Here they are considered to represent the power production of PV components. The injected power is given by the simulated scenario in the current time step t at bus i as $P_{\text{pv, i}}(t)$.

Storage

Storages are actually interchangeable with generators or loads, but intuitively they allow bidirectional power flow. Here, they represent the ESS and CS components, which, depending on the configuration, draw power from or supply power to the bus. The power injected or demanded is given by the corresponding action decided by the agent and is defined in the current time step t at bus i as $P_{\text{ess, i}}(t)$ or $P_{\text{cs, i}}(t)$ respectively.

Line

Lines connect the individual buses of the distribution grid and balance the power between them. Usually they are highly configurable, albeit here they can be considered very simplified as we are only interested in their resulting load. Therefore, they are represented only by their maximum thermal current in kilo amperes (kA).

The loading in time step t of line i is given in: [10]

$$\text{load}_i(t) = \frac{i_{\text{max},i}(t)}{i_{\text{rated},i}} \times 100 \tag{2.8}$$

In Equation (2.8), load_i represents the percentage line loading on line i in time step t. Whereas $i_{\text{max},i}$ denotes the maximum current flowing through the line, either entering or exiting, while $i_{\text{rated},i}$ refers to the line's maximum allowable current.

2.2.2 Power Flow Calculation

Since the goal of the approach is to minimize the loading on the lines in the grid, only the current $i_{\text{max},i}$ through the individual lines in the network needs to be determined in each time step in order to optimize power demand accordingly.

As only considering active power in the simulated grid and within the evaluated scenarios exclusively radial grids [11] are observed, the power flow theory is therefore limited to calculating the line loading load_i of each individual line i, by basic application of Ohm's Law ($V = I \cdot R$), and the power equation ($P = V \cdot I$). The

implementation using pandapower for modeling the power flow would allow much more sophisticated analysis of the power flow, taking into account the capabilities it contains, reactive power, phase shifting, and more in-depth analysis could be derived.

The first step is to calculate the net power at each bus i at each time step t. It can be defined as the sum of all power injections minus the sum of all power consumptions.

$$P_{\text{bus},i}(t) = P_{\text{pv},i}(t) + P_{\text{ess},i}(t) - P_{\text{load},i}(t) - P_{\text{cs},i}(t) \tag{2.9}$$

where:

- $P_{\text{pv},i}(t)$ is the power injected by PV-component of bus i at t.
- $P_{\text{ess},i}(t)$ is the power injected or drawn by the ESS-component at bus i, depending on behavior at t.
- $P_{\text{load},i}(t)$ is the power consumption of load-component at bus i at t.
- $P_{\text{cs},i}(t)$ is the power injected or drawn by the CS-component at bus i, depending on behavior at t.

The current I flowing from bus i to bus j can be estimated using the voltage V and the net power difference between the buses:

$$I_{i \to j}(t) = \frac{P_{\text{bus},i}(t) - P_{\text{bus},j}(t)}{V} \tag{2.10}$$

where:

- $I_{i \to j}(t)$ is the current flowing from bus i to bus j at time t.
- $P_{\text{bus},i}(t)$ and $P_{\text{bus},j}(t)$ are the net powers at bus i and bus j, respectively.
- V is the system voltage, assumed to be constant across the network.

The line loading $\text{load}_i(t)$ as a percentage can be expressed in terms of the maximum current $i_{\text{max},i}(t)$ through the line and the line's rated capacity $i_{\text{rated},i}$:

$$\text{load}_i(t) = \frac{i_{\text{max},i}(t)}{i_{\text{rated},i}} \times 100 \tag{2.11}$$

where:

- $i_{\text{max},i}(t)$ is derived from the current flow in Equation (2.10)between buses.
- $i_{\text{rated},i}$ is the rated capacity of a single line.
- $n_{\text{parallel},i}$ is the number of parallel lines between the two buses, if any.

3 Methodology

Based on the problem, research questions and objectives defined in the Chapter 1, this chapter describes the methodology for developing a conceptual solution that will be implemented in the further stages of the work. In Section 3.1, a conceptual approach is first designed and developed to meet the requirements and objectives. This is an analysis of the choice and composition of the circumstances and conditions under which the optimization should take place, and the choice of the components that the model should contain. From this, the necessary infrastructure and assumptions of the model can be derived. These decisions are then used in Section 3.2 discusses the data foundation required by the developed approach and how to construct and prepare it. Considerations are made in Section 4.1 to derive the necessary underlying software architecture. This is followed by a discussion of the implementation choices made in Section 4.2. The following chapter then details the implemented application of the methodology in the developed environment.

3.1 Research Design and Conceptual Framework

In order to determine the design of the study, it must first be clarified what exactly the goal is. Basically, a decision support approach is to be developed and implemented that enables the optimization of EV charging schedules in the context of V2G with the aim of a more economical and ecologically efficient use of energy - surrounded by realistic scenarios. Therefore, the charging stations need to be embedded in and interact with their usual locations. These could be residential buildings, industrial sites, or stand-alone charging sites. To make it more realistic, it should also be possible to include and consider the underlying distribution network in the optimization.

The first step is to define the scope of the optimization. It makes sense to optimize charging plans on a one-day basis, it is a popular time horizon and allowing good comparability of individual days anyway. It is also important to provide charging decisions with a reasonable resolution within one day - a compromise between realistic and practicable applicability of the results to the user's everyday life, availability of the necessary underlying data and, of course, consideration of the computational effort. This results in a compromise of 15-minute time periods and thus a division of a day into 96 time steps. This provides the scope to embed the approach in a discrete time series model.

The developed approach should be able to embed CSs in realistic scenarios. The scenarios should be as close to reality as possible. Since the reality and especially the future reality in terms of the composition of distribution grids differs to a large extent in technological equipment, general conditions and local circumstances, it should be possible to consider different initial situations as flexibly as possible.

These requirements demand the ability to design and apply scenarios that are as flexible. This raises the question of how a (theoretical) reality can be represented as accurately as possible in relation to the requirements.

3.1.1 Overview

The before mentioned forms the conceptual basis for the design of the environment to be optimized. A good representation of distribution networks with their components and their dynamics is essential. A distribution network can be thought of as a collection of individual nodes, each of which can draw power from or feed power into the distribution network. Each node can be a household, a commercial or industrial site, or even a single technology such as stand-alone PV systems, individual loads, public charging stations, or stand-alone storage systems. Of course, combinations of these are very common. A residential or industrial building with load, PV, CS and ESS. Or any other combination imaginable, such as charging stations with battery backup and possibly PV. Taking into account the interconnection of busses by lines, the collection of nodes with their individual components allows for an approximate representation of distribution networks. Distribution networks are typically connected to the medium-voltage grid at one or more of

their nodes, through which they can exchange excess or deficient power with the transmission grid. Being connected to and exchanging power with the surrounding grid, a realistic model for buying and selling energy requires the involvement of the market. For this purpose, the use of the real market data is to be considered.

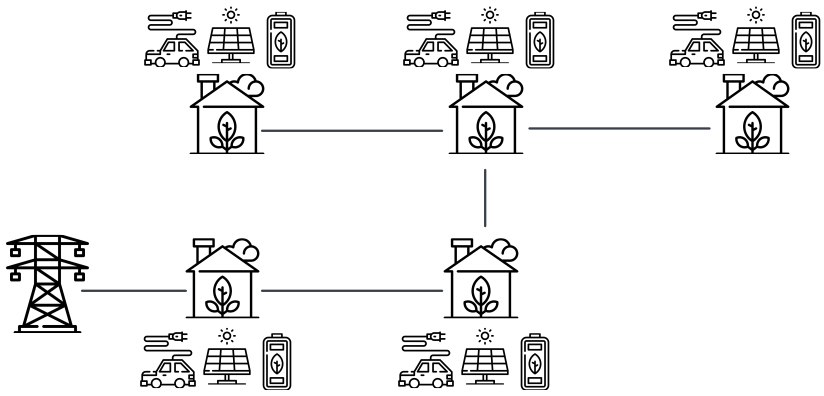


Fig. 3.1: Schematic Overview of an Example Network.

Figure 3.1 shows a schematic distribution grid of 5 buses, each representing a household equally equipped with all available technologies. The first bus is connected to the surrounding medium-voltage grid. It only serves as an example; scenarios are intended to represent any number of buses with any equipment and configuration.

In order to realistically represent the individual technologies and thus achieve a good representation of the dynamics within the grid, the approach should be based on a realistic data foundation. In particular, realistic demand and production profiles of the load- and PV- components belonging to the network are required. The load is assumed to be conventional, i.e. not intelligently controllable, which is also the nature of PV, meaning that these components cannot be influenced within a scenario, i.e. they cannot be optimized and are determined by the selected scenario – for each PV-component the peak power is configurable – and the underlying data.

As the framework to embed the model in is in place, it is necessary to consider what exactly is to be optimized within the network. It is clear from the task that EV charging schedules need to be optimized taking into account their V2G-capability. Therefore, it must be possible to control and optimize the use of all available

CS-components. To reflect this realistically, their maximum (dis-)charging power and the capacity of the EV associated with the CS should be configurable, also v2g-capability of each CS is optional. From the choice of available components, it is also desirable to optimize the use of the ESSs, which provide a way to store energy even when no EV is available. The ESS-Components should also be configurable in terms of their storage capacity and the maximum power they can supply and absorb. Based on all the before mentioned in Table 3.1 the resulting configuration options for the whole system and its components are summarized.

		Description	Data
General	Buses	Total Buses	Any Number
	Ext. Grid	Bus-IDs to ext. grid	Any Bus
	Connect.	Connection of Buses	Any Buses
Bus	Load	Load connec. to Bus	Profiles from Data
	PV	PV connec. Bus	Profiles from Data
	ESS	ESS connec. to Bus	TBO.[1](capacity; P_{max})
	CS	CS connec. to Bus	TBO.[1](v2g; capacity; P_{max})

[1] To be optimized.

Tab. 3.1: Network Configuration Options.

For a comprehensive and real scenario, further initial values for existing CS and the ESS need to be introduced within the simulation. For the CS, the model requires arrival and departure times for its corresponding EVs, as well as initial *State of Charge* (SoC) value when the vehicles arrive. The existing ESS requires an initial value of its SoC at the beginning of the simulated day. The exact generation of this data is described in Section 3.2.4.

To summarize this, considering Table 3.1, it becomes clear that the scope and size of the network in its configuration results from the scenario being simulated. The underlying PV and load profiles are specified by the scenario and are based on real data. At the same time, realistic arrival and departure times of the vehicles belonging to the CS have to be considered. The modeling of these everyday situations is done on a daily basis, in 15-minute time segments, and is intended to provide a framework for optimizing the use of ESS and CS components – with or without v2g – to tune the power consumption on each bus according to economic and environmental goals, and still provide the end user with a sufficiently charged

vehicle at departure. In addition, a simplified modeling of the load flow and the resulting load on the lines of the modeled distribution network is desirable.

3.2 Training Data Collection

In the field of RL, the quality and diversity of the training dataset play a crucial role in the effectiveness and reliability of the trained agent. A comprehensive dataset, ideally including real historical data, is critical to expose the agent to a wide range of plausible scenarios, thereby enhancing its ability to make informed decisions under varying conditions. This is particularly important in applications related to energy systems, where seasonal variations significantly influence system dynamics and decision making.

To this end, the training dataset has been designed to incorporate real-world historical data, ensuring that the agent is trained on realistic and relevant scenarios. For each episode, which simulates a 24-hour day in 15-minute segments, a random date is selected from the available dataset, ranging from January 1, 2015 to November 30, 2023. This approach ensures that the agent is exposed to a wide range of seasonal variations and historical trends, thereby enriching its learning experience.

For each selected date, the underlying data includes the day-ahead market price of energy, reflecting the cost dynamics that the agent must consider as described in Section 3.2.1. In addition, Section 3.2.2 prepares a dataset of ambient data, including solar irradiance, temperature, and cloud opacity, for a predefined location over the same time period as the market prices. This provides a realistic data source for calculating the variability of the PV energy production. The load data, representing the energy demand of a bus, is also based on a yearly data set of 74 households, which will be elaborated in the corresponding Section 3.2.3. This multi-faceted approach to data collection ensures a rich and diverse training environment, allowing the agent to learn and adapt to a wide range of scenarios, ultimately leading to more robust and effective decision making strategies.

In addition, initial conditions for each charging station, detailed in Section 3.2.4, are established using time-dependent probabilities to set realistic and configurable EV arrival times, arrival SOC, and departure schedules. This approach ensures

that the simulation reflects the diverse real-world EV charging patterns that are essential for a comprehensive training environment. By capturing these dynamics, the data base supports the development of effective strategies for managing EV charging infrastructure.

3.2.1 Energy Price: Day-Ahead-Market

For representing the dynamics of the energy market in the model the focus is on the hourly day-ahead electricity prices for Germany, covering the period from January 1, 2015, to November 30, 2023. The dataset, derived from *Ember Climate's European Wholesale Electricity Price Data*[12], compiles wholesale day-ahead electricity prices from various European countries. The data specific to Germany has been extracted. Ember Climate aggregates the data by collecting and cleaning data provided by the European Network of Transmission System Operators for Electricity (ENTSO-e), ensuring a reliable and standardized dataset for analytical purposes.

With [13] a study was found analyzing the impact of different types of flexible pricing on energy consumption for German households. In particular, the use of hourly pricing based on the day-ahead price of the *European Power Exchange* (EPEX) spot market for Germany turned out to have the greatest benefit for end users. In particular, a day-ahead based tariff called *'TARIF HOURLY'* by *awattar* is mentioned.

Component	Price (Cent/kWh)
Energy	EPEX Spot + 3%
Grid Usage	10.83
Tax, Duties	5.88
Total Price	EPEX Spot + 3% + 16.71

Tab. 3.2: Composition Flexible Tariff.

The day-ahead price is used within the model with conditions of the operating competitor *awattar* [14] to calculate the cost of energy purchase. The conditions of the flexible tariff are summarized in Table 3.2.The base price of the tariff varies by location, is fixed monthly, and is therefore ignored.

When selling energy to the grid, it is simply assumed that the customer receives 0.05 €less compared to the day-ahead price at the time of sale. This is to represent the marketer's fees and taxes.

This results in:

$$\mathrm{da}_{\mathrm{adj,buy}} = \frac{\mathrm{da}}{1000} \cdot 1.03 + 0.1671\,€/\mathrm{kWh} \tag{3.1}$$

$$\mathrm{da}_{\mathrm{adj,sell}} = \frac{\mathrm{da}}{1000} - 0.05\frac{€}{\mathrm{kWh}} \tag{3.2}$$

Where da is the original day-ahead price in $\frac{€}{\mathrm{MWh}}$, and $\mathrm{da}_{\mathrm{adj,buy}}$ and $\mathrm{da}_{\mathrm{adj,sell}}$ are the energy prices used in the model for buying and selling, respectively.

An analysis of the distribution of the original day-ahead price da converted to $\frac{texte}{kWh}$ in Figure 3.2 shows that the mean price is $0.0710\frac{€}{kWh}$, with a median price of $0.0410\frac{€}{kWh}$. The standard deviation of the prices is 0.0879. In addition, the distribution has a skewness of 3.31 and a kurtosis of 17.28, indicating a significant departure from a Gaussian distribution. Looking at the days independently, we find that of the 2891 days analyzed, 1602 days show a significant deviation from the normal distribution ($p \leq 0.05$) according to the Shapiro-Wilk test. These statistical measures, especially the fact that within single days mostly a non-normal distribution is the case, suggest that the energy price data are heavily tailed and asymmetric. Therefore, min-max normalization is used for training, which is known to be less sensitive to outliers compared to other normalization strategies.

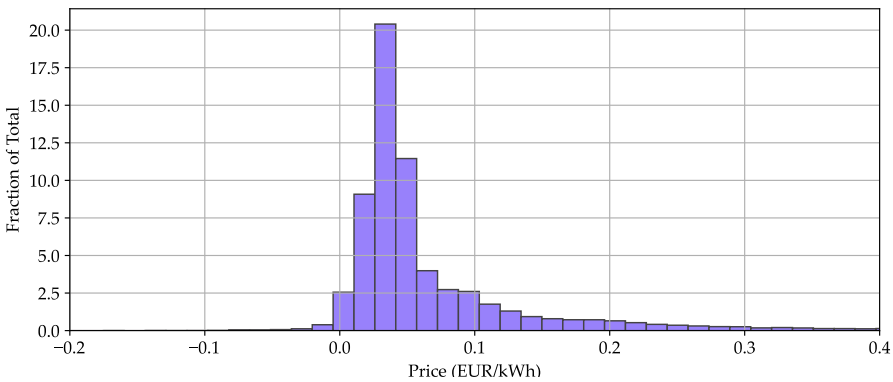


Fig. 3.2: Distribution Day Ahead form 01.01.2015 until 30.11.2023.

3.2.2 PV Profiles

The output power of a PV-system is primarily determined by the available solar radiation G and the ambient temperature. In this case *Global Tilted Irradiance* (GTI) is used. GTI represents the radiance reaching the panel's surface at a fixed tilt, inclusive of direct sunlight, sky-diffused light, and ground-reflected radiation, accounting for atmospheric conditions like cloud cover. Measured in (W/m^2), it quantifies the total solar energy available for conversion to electrical power by the PV system.

Data for solar irradiance, ambient temperature and also cloud opacity is taken from *Solcast* [15]. An API-service offering solar historical and forecasting datasets for solar irradiance and weather free for research purposes. The data is prior fetched with by for a defined location in southern Germany (Latitude: 49.293544, Longitude: 8.563727) *Solcast* recommended fixed tilt of 23 degree and prepared within the same time horizon the day ahead data was available.

The instantaneous power output from a PV array can be modeled as [16]:

$$P_{PV}(t) = P_{\text{Peak}} \left(\frac{GTI(t)}{G_{\text{standard}}} \right) - \alpha_T \left[T_c(t) - T_{\text{standard}} \right] \tag{3.3}$$

The cell temperature T_c is a function of the ambient temperature and the solar irradiance, and is given by:

$$T_c(t) = T_{\text{ambient}} + \left(\frac{T_{\text{standard}}}{800} \right) G(t) \tag{3.4}$$

In Equation (3.3), P_{Peak} represents the rated peak power of the PV system, G_{standard} is the standard solar irradiance (1000 W/m^2), and T_{standard} is the standard cell temperature 25°C. Those values are the standarized *Standard Test Conditions* (STC) values, specified in the standard.*IEC60904-3* The temperature coefficient of the PV system is denoted by $\alpha_T = (-0.004)\frac{1}{°C}$ [17]

These equations allow for the calculation of the PV system's power output at any given time, taking into account the variations in solar irradiance and ambient temperature.

Subsequently, for each PV-System existing in the network the Resulting PV Production will be calculated in each time step from its maximum Power outtake kWp and the real ambient data of irradiance, ambient temperature and the STC values. For simplification, it is assumed that all PV systems have the same tilted orientation and are localized within a narrow area due to the definition of the problem. They therefore receive the same distribution of solar irradiance for the day. As only the defined peak power of each system would therefore influence only the amplitude, but not the underlying distribution of energy production, noise is introduced for a more realistic representation of the production of different systems and a more diverse training set, which randomly corrects the production downwards for the time step by simulating clouds. The cloud opacity c, which is also available in the *Solcast* data set, is used for this purpose. Cloud opacity is given as a percentage and varies from 0% (no cloud) to 100% (full attenuation of incoming sunlight). [18] The noise is calculated as in Equation (3.5) and applied to the production power in Equation (3.6). The noise is meant to be larger for an opacity between 30% and 70% and smaller below 30% and above 70%. The values are chosen purely logically and are intended to better reflect the fact that for a fairly cloud-free or a cloud-dense time step, individual clouds or respectively clear skies between the clouds occur less frequently and therefore also the reducing effects in relation to GTI.

$$N = \begin{cases} -|\mathcal{N}(0, 0.1)|, & \text{for } 0.3 \leq c \leq 0.7 \\ -|\mathcal{N}(0, 0.05)|, & \text{otherwise} \end{cases} \tag{3.5}$$

$$P_{PV,\mathcal{N}} = \max(P_{PV} \cdot (1 + N), 0) \tag{3.6}$$

To demonstrate how typical PV profiles in a randomly chosen summer and winter day will look like in training Figure 3.3 shows the GTI of the days and their resulting PV outtake. The calculations of the preceding equations are applied an example PV-System of with a peak-power of $P_{\text{Peak}} = 15kWp$. To demonstrate the

minimizing effects on production, both the input PV-production profile P_{PV} and the noise-altered profile $P_{PV,\mathcal{N}}$ are depicted.

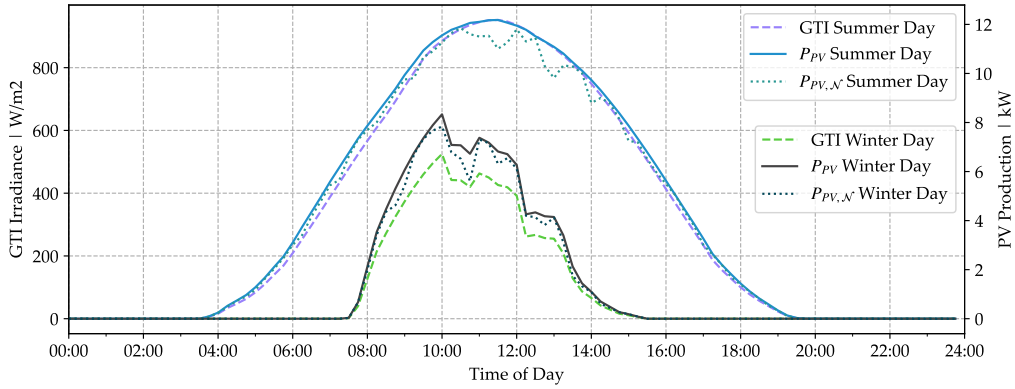


Fig. 3.3: Solar Irradiance and Example Resulting PV Prodution (with and without noise) for a $15kWp$ PV System.

3.2.3 Load Profiles

Similar to the underlying PV profiles, the load profiles used for training are taken from a data set. In [19] a dataset [20] of German single family houses was created.

The dataset is derived from a measurement campaign of the Institute for Future Energy Systems (IZES) between 2008 and 2011, in which the household electricity consumption of 497 German single-family houses was measured with a 15-minute resolution using smart meters. As mentioned above, the data are for the year 2011. Therefore, effects on load profiles due to charging of electric cars and self-use of PV production can generally be neglected. Due to its size and variability, the data set provides a good basis for robust training, as weekdays and seasonal effects are taken into account. From these, 74 profiles were selected. These profiles were gap-filled with data from similar times in the previous week to ensure continuity. In [19] the dataset was processed with a second set of measurements with additional high resolution second by second data. The longest periods in which the data are provided are in minute resolution measurements of each of the three phases.

Since this models approach uses 15 minute time steps, the minute intervals are averaged over 15 time steps. The three phases are summed for each household

and each time step, resulting in a dataset for 74 households with a resolution of 15 minutes over one year.

The resulting data of four randomly selected households for a summer and a winter day is presented in Figure 3.4

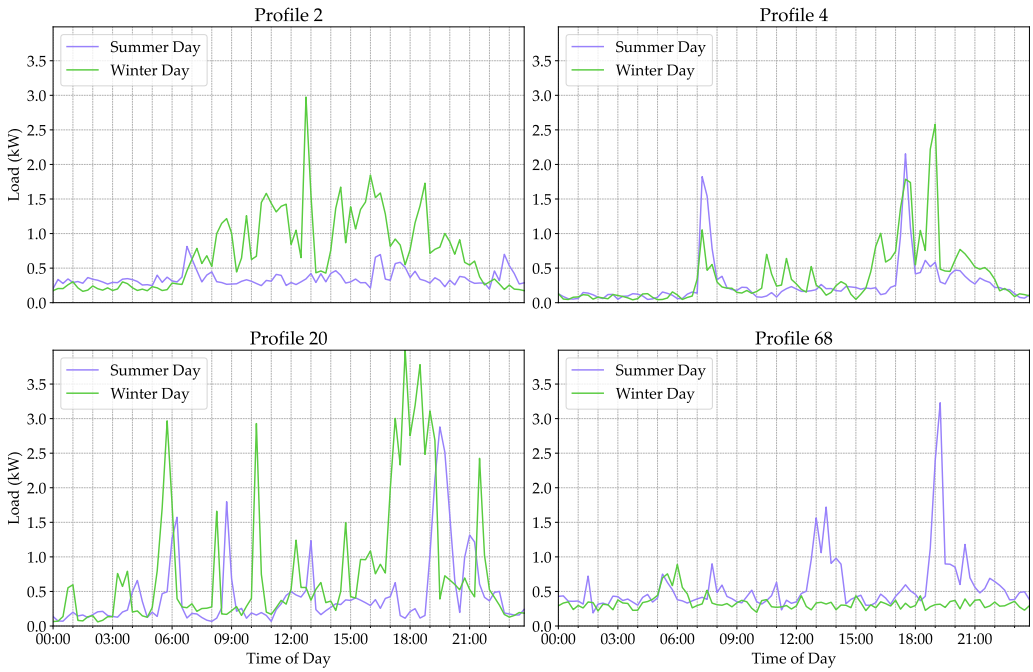


Fig. 3.4: Example Load Profiles of 4 Random Profiles and Days.

The average yearly consumption of all load profiles is 4699 kWh. Due to very different high annual consumption, profiles 8, 9, 35, 70 are dropped from the 74 profiles during training.

3.2.4 Initial Charging Station and ESS Values

CS, or their corresponding EVs, and also the ESS, require a variety of initial values for training to ensure that a diverse data base is created. This diversity is critical for the agent to generalize well across different scenarios and not overfit to a limited set of conditions. Unlike the load, PV, and day-ahead prices mentioned in the previous

subsections, which are based on real-world datasets, the input data for this case is synthesized, but an effort has been made to create a realistic data base.

For each CS in the network, at the beginning of each episode - which corresponds to a day - initial data for arrival and departure times, along with an arrival SoC, are generated to simulate the dynamics and nature of EV charging demand. This setup allows the agent to encounter and learn from a wide range of situations, from peak demand periods with multiple simultaneous charging requests to off-peak periods with minimal activity. Similarly, for the ESS, a random single value for the SoC is determined at the beginning of an episode, introducing variability in the available energy storage capacity. This randomness in the initial conditions is critical for training a robust and adaptive agent capable of making optimal decisions under varying circumstances, ultimately improving the efficiency and reliability of the charging station network.

To simulate arrival and departure times that reflect real-world conditions, time-dependent probabilities are used to realistically determine the arrival of EVs, complete with an initial SoCs, and their intended departure times. It is assumed that EV arrivals are likely to be higher during peak commuting hours, while late night charging sessions, although less frequent, remain possible. The probabilities for different time sequences are listed in Table 3.3

Time Period	Arrival Probability	Departure Probability
00:00 - 02:00	High (e.g., 0.8)	Low (e.g., 0.05)
02:00 - 07:00	Low (e.g., 0.05)	Very Low (e.g., 0.03)
07:00 - 09:00	Low (e.g., 0.05)	Very High (e.g., 0.7)
09:00 - 12:00	Medium (e.g., 0.2)	High (e.g., 0.4)
12:00 - 16:00	Medium-High (e.g., 0.3)	Medium (e.g., 0.3)
16:00 - 20:00	Very High (e.g., 0.5)	High (e.g., 0.4)
20:00 - 24:00	Medium-Low (e.g., 0.15)	Low (e.g., 0.1)

Tab. 3.3: Combined Arrival and Departure Time Probabilities.

The arrival and departure probabilities from Table 3.3 are applied in the same way to each existing charging station as shown in Algorithm 3.1. Since arrival and departure probabilities are difficult to comprehend, the resulting probability of a car being present at its charging station at each time step over the entire day is shown in Figure 3.5 based on the probabilities from the table.

In addition, realistic constraints are added to refine the resulting presence of EVs at their CS. For example, it prevents EVs from arriving just before the simulation ends to avoid scenarios where vehicles arrive without enough time to recharge. It also includes a minimum amount of time that EVs have to stay after arrival to ensure a reasonable time frame for charging sessions, mimicking real-world behavior. These elements result in a reasonably realistic and easily adaptable data foundation for training. The use of probabilities for arrival and departure also covers rare events, making the learned policy more robust.

Algorithm 3.1: Determine Presence and SoC on arrival for Cahrging Stations

```
Initialize present ← FALSE and soc ← −1;
for timestep = 0 to last_timestep − 1 do
    if not present and timestep < last_timestep − 2 then
        Determine arrival_prob based on timestep;
        present ← Random decision based on arrival_prob;
        if present then
            Set soc to a random value e.g. between 10% and 60%;
            Record timestep as arrival time;
        end
    else if present then
        if minimum presence period fulfilled then
            Determine departure_prob based on timestep;
            will_leave ← Random decision based on departure_prob;
            if will_leave then
                Record timestep as departure time;
                present ← FALSE;
                Reset soc ← −1;
            end
        end
    Update present status for the current timestep;
end
if present at last_timestep then
    Force departure and reset soc ← −1;
end
```

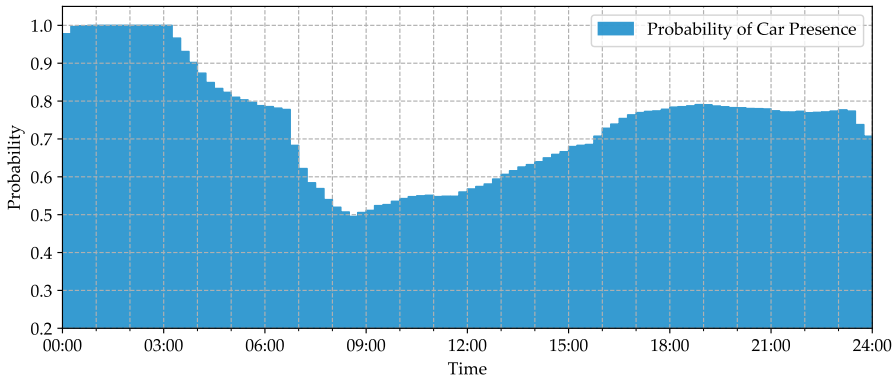


Fig. 3.5: Time series plot of charging station occupancy probability.

Initializing the underlying data for the ESSs is done in a similar, but much simpler manner. Assuming that all available ESSs in the network are always active - self-evidently limited by their physical charging and discharging capacities - only the ESS's initial SoC in the very first time step of the episode has to be randomly assigned. For this purpose, a value between 0.2 and 0.8 is randomly assigned for each SOC_{ESS_i} before the start of each episode for the first time step $t = 0$ as can be seen in Equation (3.7)

$$SOC_{ESS_i}(t = 0) \sim \mathcal{U}(20\%, 80\%) \quad \forall i \in \{1, 2, \ldots, N\} \tag{3.7}$$

3.3 Conceptual Model and Assumptions

With everything mentioned in mind, Section 3.3.1 embeds the environment to be optimized in a conceptual framework designed to achieve the defined goals, serving as an overview for the subsequent implementation Section 4.2. Prior to this, the assumptions that arise from the overall concept of the approach are specified in Section 3.3.2.

3.3.1 Model Architecture

For better understanding, Figure 3.6 shows a schematic representation based on the principles of the agent-environment interface explained in Section 2.1.2, creating a brief overview of how the environment works internally.

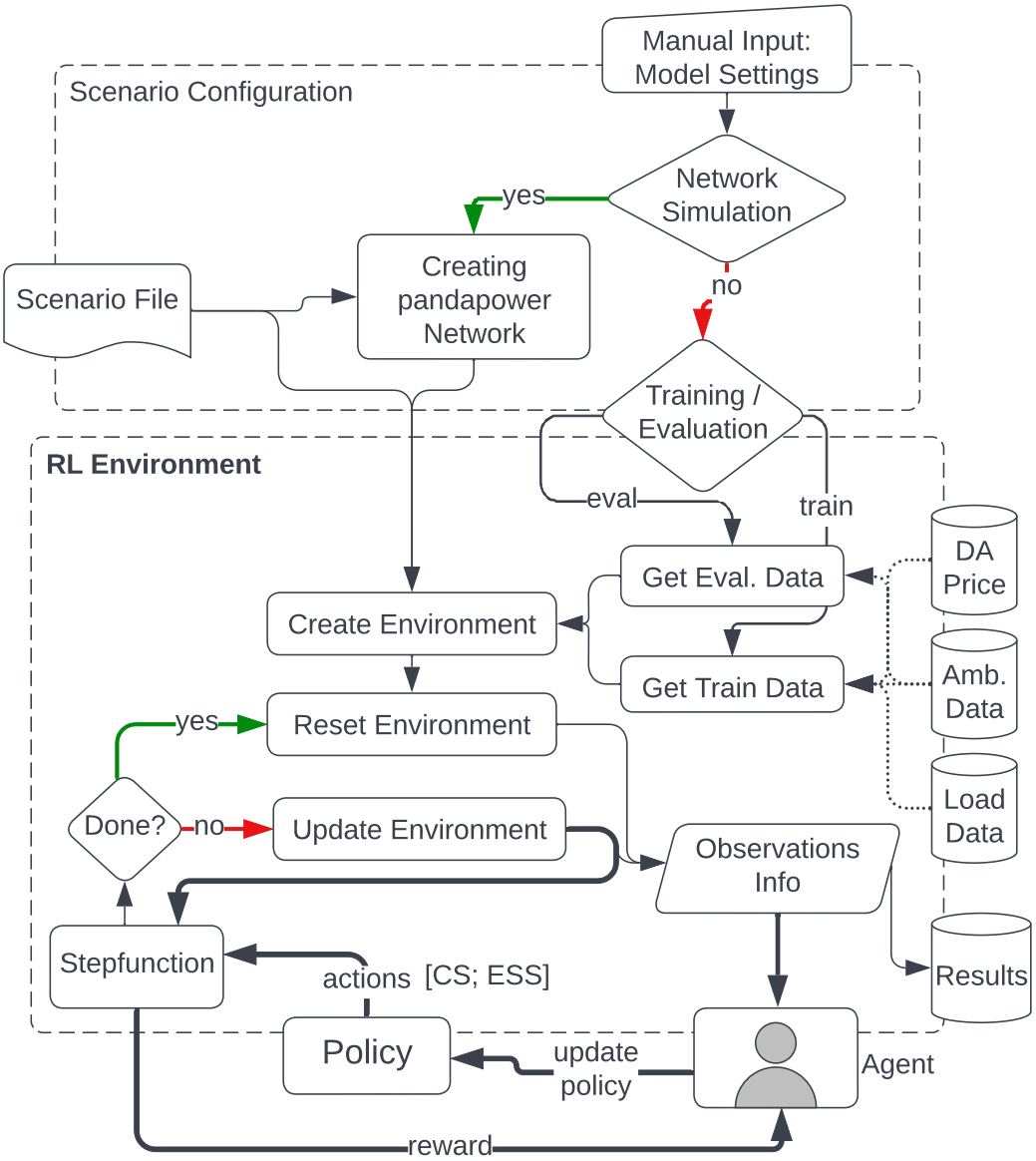


Fig. 3.6: Conceptual Overview of the Framework.

The upper box *'Scenario Configuration'* of Figure 3.6, represents the basis and initiation of the desired training or evaluation session. The training or evaluation can be configured in its shape by *Model Settings*, i.e. setting the number of episodes, algorithms used with their configuration, etc. Also, it can be decided whether the current training/evaluation should use the power flow simulation of pandapower. The *Scenario File* contains the exact configuration of the network as described in Table 3.1, available as a *.json*-file. When power flow calculation is enabled, the underlying pandapower network, with its components and the specific line interconnections between buses, is automatically created as described in Section 4.2.1.

The *'RL Environment'* box shows the environment in which the simulation will take place. Depending on whether it is a training or evaluation session, the appropriate data set is loaded into the model. On this basis, the environment to be optimized can be created; in this step, the configuration of the scenario is also taken from the *'Scenario File'*. Together with the retrieved overall data – to create the profiles within the whole training session –, and the configuration of the network, the reset-function is called. The reset, described in Section 5.3.2, loads all the required data of the simulated day / current episode – so the initial state is given. This is now observed by the agent before starting the first episode. Since there is yet no knowledge about the dynamics of the actions and the resulting reward, the first choice of actions will be very random. Nevertheless, the chosen action is fed into the Step-Function from Section 5.3.3, whereupon, since the episode is not yet *'done'*, the Step-Function calls *'Update Environment Called'*. The action of the time step is applied to the current state of the environment as described in Section 5.1, resulting in a new state. This is returned to the agent as *'Observations'*. In addition, the resulting states of episodes are stored periodically using *'Info'* into *'Results'* to gain a better understanding; in the case of evaluation, this is done for every episode in the evaluation set. Additionally, the received action is evaluated in the step function based on the new state as described in Section 5.2.3 and communicated to the agent as a reward. The agent now considers the feedback to its previous action, which was based on the last observation, and can draw conclusions about which decision might be appropriate for which state of the environment. At the end of each episode / simulated day – as soon as the last time step of 96, 15-minute intervals is reached–, the decision in the Step-Function is evaluated as *'Done'*, whereupon the environment is reset for a new episode. This continues until the configurable

amount of simulation steps has been reached, chosen depending on the complexity of the scenario to be simulated.

3.3.2 Assumptions

Assumption 1: Load profiles vary for each bus.
Assumption 2: ESS and CS possess the same maximum power for charging and discharging, further their storage is ideal; losses are not considered.
Assumption 3: EVs can arrive fast with low SoC again
Assumption 4: Agent knows when EV is going to depart.
Assumption 5: Each bus can be equipped with every technology only once.
Assumption 6: All PV profiles are based on the same irradiance data and differ only in through configured peak power and applied noise.
Assumption 7: SoC of ESS and CS are not taken into the next day
Assumption 8: Distribution of ESS and CS inital SoC and arrival /ddeparture probabilties do not vary between days

4 Implementation

Based on the concept, this chapter explains in Section 4.1 which software platforms are suitable and can be used. In addition, some implementation considerations are made in Section 4.2.

4.1 Software / Platform Selection and Rationale

In order to fulfill the aforementioned architecture and requirements, the choice of software tools and platforms to implement the developed methodology will be explained. Due to the good availability of packages and libraries in the domain of RL, the whole approach is implemented in *python*. This relies on a selected suite of software tools to facilitate exploration, development, and evaluation.

gymnasium

gymnasium [21], is an open source *python* library that provides a standardized API for communication between RL-algorithms and environments. This standardization facilitates the development and evaluation of RL models. It includes support for integrating custom environments, as discussed in Chapter 5, providing a platform for custom problems and scenarios essential for agent training and evaluation.

The library facilitates the creation of environments by predefining a standard structure for implementing custom environments, including functions and observation and action spaces. This allows for a straightforward implementation of optimization environments.

gymnasium offers the capability to easily switch between different learning algorithms, enabling a comparative analysis of their performance. This feature, alongside the ability to extensively modify and customize the library to track specific metrics, enhances its utility for in-depth RL research and applications.

StableBaseline

SB3 offers a collection of state-of-the-art RL-algorithms implemented in *PyTorch*[1], providing a reliable foundation for optimizing RL-problems.[22]

The algorithms available in SB3 are used to optimize the problem and represent the agent making the decision.

pandapower

pandapower is a *python* tool for power system analysis, incorporating a wide range of components like bus-, line-, storage-, load- and generator-components. It excels in static and dynamic analyses, such as power flow. Its *pandas*-based structure ensures ease of use and integration with other libraries, making it suitable for projects in power system engineering. [10]

pandapower will be used to simulate the power flow in each time step, in scenarios where line loading of the underlying network will be considered.

Tensorboard

TensorBoard is a visualization toolkit for *TensorFlow* that provides insights into training statistics, allowing for the monitoring of metrics, viewing graphs, and understanding model beahvior. It supports various types of visualizations, including scalar values, facilitating a deeper understanding of model behavior and performance. It's an essential tool for debugging and optimizing RL-problems, enhancing the interpretability of complex models. [23]

[1]PyTorch is an open source machine learning library for Python, favored for its flexibility and dynamic computational graph, facilitating deep learning research and applications.

Through implementing custom callbacks in the training setup and environment as explained in Section 4.2.2 more specific visual understanding of the composition of partial rewards as explained in Chapter 5 is possible.

4.2 Software Design and Implementation Details

Derived from the introductory research question from Section 1.4 and the conceptual solution approach from Section 3.1, some consequential design decisions arise from a software implementation perspective. In the following, and especially in the larger context of embedding the approach in a reinforcement learning environment in the following Chapter 5, these will be clarified and made logical.

The previous goal from the introduction calls for a freely configurable setup, which should be modular in structure, so that the model can be easily adapted, scaled, and trained in a variety of different scenarios. In this way, the resulting approach can support decision-making in environments of reality-based representations of existing small-scale distribution grids, but also allow easy scaling to larger and more complex systems and experimentation with different future technology expansion stages. In Section 3.1, it was decided for the model that for an approximately realistic representation of different households with individually configurable technological configurations in a network-integrated approach, following should apply.

The network itself is viewed as a bus system, with each bus or node essentially representing a household. The collection of all buses, considering their interconnections, can be understood as a small-scale distribution grid. Each bus can be equipped with different components, which can be configured individually. Available components are loads, PV-systems and with CSs and ESSs two controllable units. Loads, as explained in Section 3.2.3, reflect typical household load profiles over a year. The abstraction of the configurable PV system is intended to reflect typical supply profiles for different days over a year. The operation of CSs and ESS shall be optimized by the model to serve the overall system.

The option of configuring each household, i.e. each bus, individually, results in a multitude of possible scenarios. Adding each individual component to a bus is optional. As such, buses that feature single or multiple components, without

representing a household, are also conceivable. These could be detached PV systems, individual loads, public charging stations, stand-alone storage systems, or corresponding combinations, for example charging stations with battery backup and, potentially, PV systems. In the current implementation, it is noteworthy that each bus can be equipped with every technology only once, changing this would be a desirable adaptation for the representation of public charging sites with multiple charging points, for example.

The classic representation of distribution grids through buses creates a hierarchical system. Buses are subordinated to the entire distribution network resp. the overall environment. At the next level, components are subordinate to their bus according to its configuration. This aspect is transferred to the implementation. The overall environment, which is described in more detail in the following Chapter 5 from a reinforcement learning and computational point of view, is superior to the scenario-specific set of buses.

Consequently, the implemented model consists of different classes. The parent class is the environment to be optimized, basically the distribution grid. This is where information relevant to the overall system is stored and calculations on system level are performed. The superordinated environment class contains objects of the individual buses. In order to consider each bus as a separate object, a class is also introduced for this. Information and internal methods that are only relevant for the bus itself are stored and described in its object.

By introducing classes for each individual technological component, separate objects of each component can be assigned to its bus. The individual component objects store their configuration, temporal values during the simulation and define component-specific methods, e.g. for loading episode data and calculations related to the technology.

All classes used for the developed model are described hereafter.

Environment-Class: OptV2GEnv

The environment class *OptV2GEnv* is the highest level class that is organized according to the required design specifications of the RL-Framework *gymasium*.

Since it acts as the central unit of the entire model and thus also serves as a representation of the distribution grid, Chapter 5 is dedicated to it. For the sake of completeness, it is only mentioned here briefly, as it serves as the center of simulation and organization for all classes defined below.

Component: Bus-Class

Following the environment class the *Bus-Class* is the most comprehensive one. The first role entails storing the bus-specific configuration which is defined for each bus in the scenario. This includes knowledge over the existence of bus-assigned components, i.e. PV, load, the availability of a charging station and an energy storage system. Simple flags are utilized for this purpose. If a bus is in possession of a specific component, the corresponding component object is instantiated as part of the bus. Details on this can be found in the component class specific, following sections.

In addition, when instantiating the individual buses, temporal variables are initiated for the upcoming simulation to store a variety of values for the currently simulated time step. These include values relating to the power that the bus draws from the grid, resp. it feeds into the grid and the thus resulting net power. However, there are also more specific time step values, such as the exceeding power a potential charging station demands, in case the bus' components cannot provide the demand themselves, but the bus must draw from the grid. Also, the power of remaining PV-producton being not consumed by any of the bus components is stored. In addition, the energy costs respectively the revenue of the current time step will be stored.

These values are used either for the ongoing simulation, the calculation of power balances or for the reward and penalty calculation at each time step describes in Section 5.2.3.

In addition, the bus class defines methods that are called at the start of each episode in order to pass on the retrieval of the PV- and load-profile for the simulated day to the respective subordinate component object method.

Component: Load-Class

Objects of the *Load-Class* are only instantiated by a parent Bus-Object if it has a Load-Component. The Load is bound to its bus. Temporal variables are introduced to store the load profile over an episode and the power consumed in the current time step of the simulation.

Furthermore, the *Load-Class* defines a method that is called by its bus at the start of each episode and retrieves the load profile and stores it in its Load instance.

Component: PV-Class

An instance of the *PV-Class* can also only be instantiated by a Bus-Object - provided it possesses a PV component. In the configuration of the scenario the individual PVs peak power is defined, and will be stored in the corresponding PV-object. In addition, the fixed constants for calculating the output power are also set at initialization for later use as described in Section 3.2.2. In the current setup, the constants are the same for all PV instances. For future modifications, it is desirable to be able to assign different values for each individual PV instance, as this will allow easy expansion to include more individual parameters, allowing more realistic and distinct production profiles, e.g. by introducing the degree of inclination, etc.

The only method defined in the *PV-Class* is for calculating the PV profile of the episodes, i.e. the output power at each time step over the entire episode. It is called at the beginning of each episode by its parent bus object. As described in Section 3.2.2, the calculation of the individual PV profiles is based on real historical time-series ambient data of solar irradiance, temperature and cloud cover over the current simulated day in addition to the specific peak power of the PV. The ambient data of each episode is globally defined and valid for all PV instances.

As already mentioned, for PV systems with the same peak power, this results in very similar profiles within an episode, although they may all differ slightly due to cloud cover noise.

Component: CS-Class

An instance of the *CS-Class* can also be instantiated exclusively by a bus object, provided that it has a CS component. The specification of the individual charging station is also taken from the scenario configuration. At initialization, parameters about the availability of V2G technology, implemented as a flag, the maximum (dis-)charging power and the battery capacity of the EV belonging to the charging station are stored.

In addition, variables required for the episodes and time steps during the simulation are introduced. These are the action value determined by the agent for this CS, the state whether the car is at the charging station, realized by a flag, and the current SoC.

The main method of the *CS-Class* is to calculate the (dis-)charging power of the current time step, based on its current action value. How the action is applied is explained in Section 5.2.2. To ensure compliance with the physical limits of the charging process, this method relies on two functions that also belong to the charging station object.

Component: ESS-Class

Again, an ESS object can only be initiated as part of a Bus-object provided it possesses a PV component. The ESS class is nearly identical to the CS-Class. The sole difference is that an ESS is always available, as it cannot leave, and unlike for a non-V2G capable CS-Component, the discharge capability is always accessible. An object of the ESS class therefore only has the configuration parameters specified in the scenario for capacity and maximum (discharge) charging power.

During the simulation, variables for the selected action of the current time step specific to the ESS, the resulting charging power and the current SoC can be accessed.

The methods and functions of the ESS class are also linked to those of the CS class – calculating the current charging power derived from its action.

4.2.1 Integration with Power Network Calculations

As mentioned in the methodology, it should also be possible taking into account the power flow of the underlying network of the environment in its different scenario configurations. At least to determine the resulting loading of the lines. For this purpose, an pandaopwer schematic corresponding to the scenario must be generated from the existing scenario configuration as given in Table 3.1.

For this purpose, a module named *create_pp_network* has been added to the repository, which automatically creates a panda power schematic of the network from the scenario configuration *.json* file. The scenario configuration contains the number of buses, a representation of the bus connections and the individual equipment of each bus. The bus connections are given by a symmetric matrix, as in Equation (4.1), specifying which bus is connected to which, the diagonal is only an aid connecting each bus to itself. Consequently, for each bus in the scenario, a bus is created in the representative pandapower topology, to which the elements described in Section 2.2.1 are then connected, depending on the configuration. Adding these elements is only important to have them available, no underlying data is required, because – in case activated for training or evaluation – the power flow calculation is performed for each call of the step function, i.e. for each time step. Thus, the current power data of each component in each time step can be loaded directly from the environment into the pandapower representation of the network. Then, the power flow calculation for this time step can be performed internally by the pandapower module. [10]

$$\mathrm{Bus}_{\mathrm{con}} = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{pmatrix} \tag{4.1}$$

4.2.2 Tracking of Partial Penalties and Rewards

In the nature of choosing a single objective is the poor interpretability of the partial rewards and penalties that compose the overall reward signal. For this reason, the

built-in feature of Stable-Baselines3 of intgrating custom callbacks was utilized in combination with Tensorboard. A custom callback is initiated when configuring the training, which receives a dictionary containing the individual partial rewards at the end of each step of an episode via the return of the step function. These are averaged in the same way as the standard reward in Tensorlog and can therefore be interpreted visually in the Tensorboard dashboard alongside the overall reward. This makes it much easier to understand the training process.

5 OptCharge Reinforcement Learning Environment

The main part of the approach developed in this thesis is dedicated to the Reinforcement Learning Environment. It provides the framework for the interaction of the individual, previously described, conceptual parts at the simulation level within the training and subsequent evaluation of the models of different scenarios.

The following part is mainly dedicated to how the conceptual approach can be integrated into an environment whose structure is partly given by the RL framework *gymnasium*. But also the way how to handle different data to make the simulation achieve its intentions is covered. The goal is to provide a comprehensive description starting from the underlying database and model configurations, through the individual simulation steps, to the trained model and its evaluation.

5.1 Network Operation Logic

For the model to work reliably, calculations must be performed in the environment at each time step to compute the new state from the previous state and to provide the agent with the corresponding observations. These calculations are described in the following.

When calculating the power balance at time step t for each bus, the supply or dissipation of each component belonging to the bus must be taken into account. The total net power of the bus is calculated as follows:

$$P_{\text{net},i}(t) = P_{\text{load},i}(t) + P_{\text{cs},i}(t) + P_{\text{ess},i}(t) - P_{\text{pv},i}(t) \tag{5.1}$$

Whereas:

- $P_{\text{load},i}(t)$: represents power demand of Load-Component at bus i at time t, derived from the current episodes load profile of the bus
- $P_{\text{cs},i}(t)$: represents power demand / supply of CS-Component at bus i and time t, calculated from the CS-Action $a_{\text{cs},t}$ in Section 5.2.2
- $P_{\text{ess},i}(t)$: represents power demand / supply of ESS-Component at bus i and time t, calculated from the ESS-Action $a_{\text{ess},t}$ in Section 5.2.2
- $P_{\text{pv},i}(t)$: represents power supply of PV-Component at bus i and time t, derived from the current episodes load profile of the bus

If a bus does not possess one of these components, the respective X-Component refers to $P_{\text{X}} = 0$ in the current time step t.

From Equation (5.1) also is calculated :

$$P_{\text{from grid},i}(t) = \max(0, P_{\text{net},i}(t)) \tag{5.2}$$

$$P_{\text{to grid},i}(t) = \min(0, P_{\text{net},i}(t)) \tag{5.3}$$

and the unused PV-production:

$$P_{\text{pv, avail},i}(t) = \max\Big(0, P_{\text{pv},i}(t) - \Big(P_{\text{load},i}(t) + \max(0, P_{\text{cs},i}(t)) + \max(0, P_{\text{ess},i}(t))\Big)\Big) \tag{5.4}$$

5.2 State, Action, and Reward Definitions

The Theory Section 2.1.2 explained how agent-environment interactions in an RL-optimization problem work in principle. Section 3.1 showed how this applies to the specific problem. The following section focuses on the state, actions, and observations specified for our problem. In the same way as the agent-environment communication is chronologically staggered, states of the environment are described

first, followed by the resulting actions, and finally the design and computation of the composed overall reward signal consisting of partial penalties and rewards is presented. In the last part, the weighting and effects of the partial rewards are briefly discussed.

5.2.1 States

The design of the approach prescribes that the size of the observation space varies depending on the scenario being simulated. At the core, there's a global observation that captures the day-ahead price applicable to the entire environment. In addition, the observation space expands with the complexity of the scenario, as each component – be it load-, PV-, CS-, or ESS-components – is monitored individually. For observations of energy prices and PV-components, it is possible to flexibly determine how many future time steps can be considered, essentially acting as a forecast. This reflects the real world, where day-ahead prices are known in advance for the entire day, and PV output can be reliably predicted using weather and environmental data. For other types of observations, only the current time step is considered for simplicity. The following sections provide a detailed breakdown of each observation component and how they integrate into the overall model.

Observation: Day-Ahead Price

Including information on *Day-Ahead* (DA)-prices is essential for enabling cost-effective energy management decisions within the environment. By default, the observation includes the DA price of the current time step. The extent to which future time steps are observed is determined by the prediction horizon $n_{\text{pred_da}}$. The observations corresponding to energy prices are outlined as follows:

Day-Ahead Price Observation ($\mathbf{obs_{da}}$): This is an array that equips the agent with the current and a chosen number of future energy prices, represented by $n_{\text{pred_da}}$.

The formulation of the day-ahead price observation is expressed through:

$$\text{obs}_{\text{da}}(t) = \{\text{da}_{\text{ep}}(t), \text{da}_{\text{ep}}(t+1), \ldots, \text{da}_{\text{ep}}(t + n_{\text{pred_da}})\} \tag{5.5}$$

Here, $\text{obs}_{\text{da}}(t)$ denotes the array of observations, $\text{da}_{\text{ep}}(t)$ represents the normalized episodes day-ahead price at the current time step t.

As detailed in Section 3.2.1, DA-prices are normalized using min-max normalization at the start of each episode. This ensures that the agent is attuned to relative changes in energy prices and that decision-making is focused on relative changes rather than being distracted by their absolute values, which can vary widely between episodes. Access to immediate and short-term forecasts of energy prices enables the agent to anticipate cost fluctuations and develop strategic plans accordingly. This enhances the agent's ability of improving the financial efficiency of the system in response to the dynamic nature of energy markets.

Observation: Bus-Loads

Knowledge of the load demand within the distribution network is critical for informed decision making. Therefore, observations of all load components within the simulated scenario are provided. These observations are described as follows:

Bus-Load Observation (obs_{load}): This array provides normalized values representing the electrical load demand at each bus within the network for the current time step.

Normalization of the load values is achieved by dividing the actual load by a predefined maximum value $P_{\text{load, max}}$ = 5kW, as 2597303 of 2600064 time step power values in the data of Section 3.2.3 are < 5 kW, that is more than 99 % of all values. This ensures a consistent scale for the observed data across the network. The Load-Observation is captured in the following equation:

$$\text{obs}_{\text{load}}(t) = \left\{ \frac{P_{\text{load},b}(t)}{P_{\text{load, max}}} \middle| b \in \text{Buses with Load} \right\} \tag{5.6}$$

Here, $\text{obs}_{\text{load}}(t)$ represents the immediate load demand at each bus, providing information of the current energy consumption across all buses.

The observation of normalized load data equips the agent with a clear understanding of the distribution and intensity of power demand. This is pivotal for making

informed decisions regarding energy allocation, balancing supply and demand, load management of controllable components, and enhancing the operational efficiency within the network.

Observation: Bus PV-Production

Observing the PV-production at each bus connected to a PV-system is crucial for the agent's decision-making, especially for integrating renewable energy into the charging station operations. These observations are structured as follows:

PV-Production Observation ($\mathbf{obs_{pv}}$): This array provides the agent with normalized values of solar energy production at each bus for the current time step and an adjustable number of future time steps, determined by the prediction horizon $n_{\text{pred_pv}}$. The values are normalized with respect to the maximum value $P_{\text{pv, max}}$ of the PV-component with the highest kWp in the environment, ensuring uniform scaling across the network. The structure of this observation is elaborated with:

$$\text{obs}_{\text{pv}}(t) = \bigcup_{i \in \text{Buses with PV}} \left\{ \frac{P_{\text{pv},i}(t+j)}{P_{\text{pv, max}}} \middle| j = 0, 1, \ldots, n_{\text{pred_pv}} \right\} \tag{5.7}$$

In this equation, obs_{pv} denotes the aggregated solar energy production observations for all buses equipped with PV systems. $P_{\text{pv},i}(t+j)$ specifies the PV power output for bus i at time step $t+j$, where j spans from 0 to $n_{\text{pred_pv}}$, including both the current time step and a specified number of future steps.

These observations offer insights into the current and near-future availability of solar power, aiding the agent in making informed decisions about harnessing renewable energy for charging or feed-in operations. The normalization of PV production to uniform scaling across the networks will allow the agent to get a better understanding of the influence of those values.

Observation: Charging Station

The decision-making process of the agent is significantly enhanced by observations that detail the operational status of each charging station and the vehicles present at discrete time intervals. These observations are structured as follows:

Charging Station Occupancy and EV Observation ($\mathbf{obs_{cs}}$): For every charging station $cs \in \{1, \ldots, n_{\text{cs}}\}$, an array $\text{obs}_{\text{cs},i}$ provides insights into:

- The not occupied status, indicating with a sequence $\{0, 0, 0\}$ the absence of a car at time t, thus marking the station as vacant.

- The presence of an EV through a sequence $\{1, \frac{t_{\text{dep,cs}}-t}{T_{\text{last}}}, \text{SoC}_{\text{cs}}(t)\}$, where the first entry 1 observes the presence of an EV, $t_{\text{dep,cs}}$ represents the normalized departure time of the car at station cs – whereas T_{last} is the last episodes time step –, and $\text{SoC}_{\text{cs}}(t)$ its state of charge at time t. This conveys the presence, the remaining time until the car departs and its current charge level.

For each charging station $cs \in \{1, \ldots, n_{\text{cs}}\}$:

$$\begin{aligned} \text{obs}_{\text{cs},i}(t) &= \begin{cases} \{0, 0, 0\}, & \text{if no car is present at } cs \text{ at time } t, \\ \{1, \frac{t_{\text{dep,cs}}-t}{T_{\text{last}}}, \text{SoC}_{\text{cs}}(t)\}, & \text{if a car is present and } t < t_{\text{dep,cs}}. \end{cases} \\ \text{obs}_{\text{cs}}(t) &= \bigcup_{i=1}^{n_{\text{cs}}} \text{obs}_{\text{cs},i}(t) \end{aligned} \tag{5.8}$$

The collective observation obs_{cs} combines these individual arrays $\text{obs}_{\text{cs},i}$ for each station, providing a comprehensive view of the state of the networks charging infrastructure.

Through these observations, the agent can assess both the current and expected state of each charging station, which is critical for orchestrating efficient (dis-)charging strategies and prioritizing EVs that are about to leave.

Observation: Energy Storage System

Knowing the states of the available ESSs in the network is critical for good decision making. Therefore, observations about the SOC are provided for each ESS. These observations are structured as follows.

ESS State of Charge Observation ($\mathbf{obs_{ess}}$): This array captures the current SoC for each ESS associated with the buses in the network, providing a snapshot of the available energy capacity at each location.

The SoC values are observed directly without the need for normalization, given the bounded nature of SoC values. The structure of these observations is detailed in the following equation:

$$\text{obs}_{\text{ess}}(t) = \left\{ \text{SoC}_{\text{ess},b}(t) \,\middle|\, b \in \text{Buses with ESS} \right\} \tag{5.9}$$

In this context, $\text{obs}_{\text{ess}}(t)$ provides a comprehensive view of the energy storage levels across the network, with $\text{SoC}_{\text{ess},b}(t)$ reflecting the SoC of the ESS at bus b at time t.

These SoC observations empower the agent with critical insights into the energy storage landscape, facilitating optimal energy management strategies. This is essential for ensuring the effective utilization of stored energy, thus supporting the reliability and efficiency of the overall network operation.

Total Observations at times step t :

The last observation $\text{obs}_{\text{ts}}(t)$ normalizes the current time step to provide a sense of progression through the episode and thus the current time steps time of the day, calculated as:

$$\text{obs}_{\text{ts}}(t) = \frac{t}{T_{\text{last}}} \tag{5.10}$$

where t represents the current timestep and T_{last} denotes the last timestep of the episode.

All observations of the various components and general states of the environment are combined into a single observation vector obs before being returned to the agent. Reflecting many observable states of the environment at the current time step t, this provides a comprehensive foundation for informed decision making. All states presented in Equation (5.11) are normalized between 0 and 1.

$$\begin{aligned}\text{obs}(t) = \{&\text{obs}_{\text{ts}}(t), \text{obs}_{\text{cs}}(t), \text{obs}_{\text{ess}}(t), \text{obs}_{\text{pv}}(t),\\ &\text{obs}_{\text{load}}(t), \text{obs}_{\text{da}}(t)\}\end{aligned} \tag{5.11}$$

5.2.2 Actions

Similar to observations, the number of actions within a simulated scenario depends on its specific setup. As described in the concept chapter, the ensemble of controllable units is much smaller than the total number of components. In this context, only CS- and ESS-componentes are included controllable, owning the ability to influence and optimize the state of the environment. The actions of the current time step is denoted as A_t. Each bus within the network is limited to a maximum of possessing one CS- and one ESS-component. The actions are tuned to their respective components. The dimension is the sum of the number of CS- and ESS-components in the network, formulated as $n_{\text{act}} = n_{\text{cs}} + n_{\text{ess}}$. Consequently, the maximum number of actions in an environment is expressed as $n_{\text{act}} = 2 \cdot n_{\text{Bus}}$. This number is lower in scenarios characterized by fewer ESS and CS components.

The optimization algorithms presented in the Section 2.1 chapter benefit from a symmetric action spaces, therefore all actions are defined as continuous variables, which are constrained in the [−1,1] space over all n_{actions}.

Actions	number	Range
ESS-Actions	n_{cs}	[−1,1]
CS-Actions	n_{ess}	[−1,1]

Tab. 5.1: Environment Actions.

CS- and ESS-actions are essentially applied in the same way, their continuous action value is translated into power requests via the current state of the Component. However, there are subtle differences in their practical application that will be discussed in the following sections.

CS-Actions

CS-actions are symbolized by $a_{t,\text{cs}}$, where t denotes the time step, cs refers to the CS the action is applied to.

CS-components come in two different types – with and without the ability to perform V2G. For cs without V2G the applied action can only perform charging operations, with V2G it can also perform discharging operations. Therefore, for the CS-action – always within [−1,1] – it is necessary to distinguish between these cases. In addition, the physical limits of the EV battery capacity must be respected during charging and discharging. The calculation of the final power applied to each the CS by its action is performed as follows.

Algorithm 5.1: Applied Power Calculation $P_{\text{cs,t}}$ for a CS from action $a_{\text{cs},t}$

if *cs not capable of v2g* **then**
  $a_{\text{cs},t} \leftarrow \frac{a_{\text{cs},t}+1}{2}$;
$P_{\text{cs,des}} \leftarrow P_{\text{cs,max}} \cdot a_{\text{cs},t}$;
if $P_{cs,des} \geq 0$ **then**
  $C_{\text{cs,remain}} \leftarrow (1 - \text{SoC}_{\text{cs},t}) \cdot C_{\text{cs,max}}$;
  $P_{\text{cs,remain}} \leftarrow C_{\text{cs,reimain}} \cdot 4$;
  $P_{\text{cs,feas}} \leftarrow \min(P_{\text{cs,max}}, P_{\text{cs,remain}})$;
  $P_{\text{cs,t}} \leftarrow \min(P_{\text{cs,des}}, P_{\text{cs,feas}})$
else
  $C_{\text{cs, avail}} \leftarrow \text{SoC}_{\text{cs},t} \cdot C_{\text{cs,max}}$;
  $P_{\text{cs,avail}} \leftarrow C_{\text{cs,avail}} \cdot 4$;
  $P_{\text{cs,feas}} \leftarrow \min(P_{\text{cs,max}}, P_{\text{cs,avail}})$;
  $P_{\text{cs,t}} \leftarrow \max(P_{\text{cs,des}}, -P_{\text{cs,feas}})$
end

Algorithm 5.1 outlines the calculation of the power applied $P_{\text{cs,t}}$ to each CS based on the action $a_{t,\text{cs}}$ at time step t. It takes into account whether the CS is capable of V2G operations and adapts the action accordingly. For CS without V2G, the action is adjusted to ensure that only charging will be available. The desired power $P_{\text{cs,des}}$ is then determined proportional to the maximum power $P_{\text{cs,max}}$ the CS is capable of charging and discharging, based on the action. $P_{\text{cs,max}}$ is scenario dependent and can be configured for each CS individually. Depending on whether the desired

power is for charging or discharging, the remaining or available capacity of the CS is calculated, taking into account the state of charge (SoC) and the maximum capacity $C_{\text{cs,max}}$ – also configurable individually. From this by multiplying $C_{\text{cs,remain}}$ by 4, $P_{\text{cs,remain}}$ is calculated – necessary because of the need to transfer the kWh value to the 15-minute time domain. The feasible power $P_{\text{cs,feas}}$ shows how much power could theoretically be applied based on the remaining or available headroom capacity. Since this tends to exceed the maximum power of cs, $P_{\text{cs,feas}}$ and $P_{\text{cs,max}}$ are compared. Finally, the power applied at time t, $P_{\text{cs,t}}$, is a feasible value either for charging or discharging, ensuring that the physical limits of the EV-battery are respected.

ESS-Actions

ESS-actions are applied analogously to the CS-actions, with their respective configuration. Only the distinction between the ability to charge or discharge and the corresponding adaptation of the action is omitted. As with CS, for ESS the desired charging or discharging request $P_{\text{ess,des}}$ of the action is constrained based on their current state and taking into account physical limits, resulting in the power value $P_{\text{ess,t}}$.

5.2.3 Penalties and Rewards

This section breaks down the composition of the reward function, essential for guiding the actions from Section 5.2.2, chosen based on the observations in Section 5.2.1, within the model. The reward-signal is designed of various partial penalties and rewards that compose a single scalar reward signal. Through weighting the individual parts the objective can be balanced in different directions to encourage wanted and discourage unwanted behavior, enabling to achieve different goals for the same scenario.

The following begins with a brief overview of the individual partial signals, their purpose and how they contribute to the system's overall objectives and dynamics. In the Appendix B there's also briefly some experimental penalty signal mentioned, which is not part in the later evaluated scenarios, but helped during the process to understand the models dynamic.

Penalty: Buying Energy from Grid

The penalty for buying energy from the grid is the most logical to incentivize the agent to manage energy demand economically efficient. Through design a low penalty for buying energy from the grid when prices are relatively low and high penalty when prices are relatively high, is applied. The penalty p_{grid} is formulated to reflect the dynamic nature of energy prices, taking into account the normalized DA-price at time t, denoted as $\text{da}_{\text{ep}}(t)$, and the energy price range within the episode, $\text{da}_{\text{ep,rg}}$. The total energy drawn from the grid is accumulated across all buses. Naturally, there is a difference between looking at each bus individually and looking at the entire network as a whole. If only the total network balance is considered, the monitored consumption would be lower as the buses also supply each other. However, it was decided to find optimized solution for each individual bus. The cost of purchasing energy is represented by penalty $p_{\text{grid}}(t)$ calculated as follows:

$$\begin{aligned} E_{\text{from_grid}}(t) &= \frac{1}{4}\sum_{i=1}^{\text{n}_{\text{bus}}} P_{\text{from_grid},i}(t) \\ p_{\text{grid}}(t) &= (1.03 \cdot \text{da}_{\text{ep}}(t) + 0.1671 \cdot \text{da}_{\text{ep,rg}}) \cdot E_{\text{from_grid}}(t) \end{aligned} \tag{5.12}$$

Since the power at the buses is only available over a time step of 15 minutes, it is divided by 4 to calculate the energy in kWh. Then, the flexible energy tariff consider day-ahead prices from Section 3.2.1 is applied to the normalized price of the time step and additionally scaled by the price range of the episode. This is not entirely accurate, as the offset actually has to be applied to the real price; the scaling helps to counteract this. However, the intended direction of the incentive remains clear. This adjusted normalized price ensures that buying energy is always associated with a higher penalty compared to the reward of selling in the same time step. This encourages energy conservation and the strategic use of storage or direct use of PV-production.

Reward: Selling Energy to Grid

Conversely, to the penalty of buying energy, the agent is rewarded for selling energy back to the grid. This reward $r_{\text{grid}}(t)$ is calculated as follows:

$$\begin{aligned} E_{\text{to_grid}}(t) &= \frac{1}{4}\sum_{i=1}^{\text{n}_{\text{bus}}} P_{\text{to_grid},i}(t) \\ r_{\text{grid}}(t) &= ()\text{da}_{\text{ep}}(t) - 0.05 \cdot \text{da}_{\text{ep,rg}}) \cdot E_{\text{to_grid}}(t) \end{aligned} \tag{5.13}$$

The calculation is performed in the same way as for the penalty in Equation (5.12), except that the feed-in power of the buses and a distinct adjustement applied to the current normalized day-ahead price $\text{da}_{\text{ep}}(t)$ is used – as explained in Section 3.2.1. The reward for selling energy $r_{\text{grid}}(t)$ is derived from the total feed-in energy scaled by the adjusted selling price. This mechanism encourages the agent to generate or store energy when prices are low and sell when prices peak.

By dynamically adjusting the rewards and penalties associated with energy transactions based on real-time pricing, the system fosters efficient energy management, aligning the agent's objectives with broader energy conservation and sustainability goals.

By imposing higher prices for energy purchases and offering lower returns for energy sales, the framework instills a more realistic market scenario that not only disincentives excessive energy sales but also fosters greater self-consumption.

Penalty: SoC of CS on Departure

From EV- and user- perspective an important goal is to reach a sufficient SoC on departure. Therefore, a penalty reflecting the charging state on departure of EVs is introduced.

This mechanism designed as in Equation (5.14), serves to penalize the agent for permitting EVs to depart the charging station (CS) with a SoC below an optimal charge level. The penalty p_{cs} for all EV leaving the CS a time step t is calculated as follows:

$$
\begin{aligned}
\text{Let } \mathcal{L} &= \{i \in \text{EVs} \mid \text{EV } i \text{ is leaving at time } t\} \\
p_{\text{cs},i}(t) &= ((1 - \text{SoC}_{\text{cs},i}(t+1)) \times 2)^2 \quad \forall i \in \mathcal{L} \\
p_{\text{cs}}(t) &= \sum_{i \in \mathcal{L}} p_{\text{cs},i}(t)
\end{aligned}
\tag{5.14}
$$

Where $p_{\text{cs,i}}(t)$ represents the individual penalty of EV i in the set $\mathcal{L}$ leaving in time step t, $\text{SoC}_{\text{cs},i}(t+1)$ represents the SoC of the EV in the next time step – the value of the next time step is the actual SoC charged when the EV is leaving. The difference between the ideal SoC of 1 and the actual SoC post-charging is squared and scaled by a factor of 2, to penalize larger deviations more severely, thus incentivizing strategies that prioritize full charging prior to departure. The penalty $p_{\text{cs}}(t)$ is the sum of all EVs in the set $\mathcal{L}$ leaving in this time step. It is noteworthy that penalties are not imposed at the simulation's final timestep to allow an unbiased evaluation of the system's final state. Also it will always be 0 if no EV is leaving in the current time step t

Penalty: Remaining PV-Production

To incentivize the agent to high self-consumption of the buses PV-production a penalty is introduced for any PV-production that remains unused in case an available CS-Component or the ESS could have consumed it.

This applies to the set of selected PV-Components $\mathcal{S}_{\text{PV}}$ based on:

$$
\mathcal{S}_{\text{PV}}(t) = \left\{ \text{PV}_i \;\middle|\; \begin{array}{ll} \begin{bmatrix} \text{SoC}_{\text{cs},i}(t) \neq -1, \\ \text{SoC}_{\text{cs},i}(t) \leq 0.92, \\ P_{\text{cs,max},i}(t) - 0.2 > P_{\text{cs},i}(t), \end{bmatrix} & \text{for } i \text{ with CS,} \\ \begin{bmatrix} \text{SoC}_{\text{ess},i}(t) \leq 0.92, \\ P_{\text{ess,max},i}(t) - 0.2 > P_{\text{ess},i}(t), \end{bmatrix} & \text{for } i \text{ with ESS,} \end{array} \quad \forall i \in \text{Buses with PV} \right\}
\tag{5.15}
$$

The selection criterion in Equation (5.15) ensures that, for a bus equipped with a PV component, specific conditions are met before any penalty is applied. The bus

has either to be equipped with ESS or CS. If equipped with a CS it an EV needs to be present, indicated by a State of Charge (SoC) not equal to -1. Additionally, it is ensured that there is sufficient SoC headroom (SoC $\leq$ 0.92), so can still be charged. Furthermore, the current power being supplied to the $P_{\text{cs},i}(t)$ must not be at its limit $P_{\text{cs,max},i}(t)$. The margin for the power conditions simply is some fair threshold as the agent is acting a continous space.

Similarly, if the bus is equipped with an Energy Storage System (ESS), there is ensured sufficient capacity left (SoC $\leq$ 0.92) and the current power being supplied $P_{\text{ess},i}(t)$ must not be at its limit $P_{\text{ess,max},i}$.

If either of the two combined conditions for CS or ESS applies, the PV-Component is considered for the calculation of penalty p_{pv} in:

$$p_{\text{pv}}(t) = \sum_{i \in \mathcal{S}_{\text{PV}}} \begin{cases} \frac{P_{\text{pv, avail},i}(t)}{P_{\text{pv},i}(t)} & \text{if } P_{\text{pv},i}(t) > 0 \\ 0 & \text{otherwise} \end{cases} \tag{5.16}$$

The penalty p_{pv} quantifies the wasted potential of PV-power by evaluating the fraction of unused PV-power across all buses within the set $\mathcal{S}_{\text{PV}}$ during time step t. For each bus i, the unused fraction is calculated as the ratio of the unused PV-power $P_{\text{pv, avail},i}$ to the generated PV power $P_{\text{pv},i}$. If there's no PV production at a bus ($P_{\text{pv},i}(t) = 0$), the unused fraction for that bus is considered to be 0, indicating no wasted PV potential.

The conditions within the set $\mathcal{S}_{\text{PV}}$ ensure that the penalty only is applied, if there is chance for improvement. By summing up these fractions, p_{pv} captures the overall unused PV power in the network, giving an incentive to minimize this penalty by optimizing the use of PV-components.

Penalty for Line Load

When *pandapower* load flow simulation is enabled in a training configuration, the model also offers to apply a penalty p_{line} that takes into account the line load of each line in the network. This is applied to each available line as follows:

$$p_{\text{line},i}(t) = \begin{cases} 5 & \text{if } load_i > 100\% \\ 1 & \text{if } 80\% < load_i \leq 100\% \\ 0.2 & \text{if } 50\% < load_i \leq 80\% \\ 0 & \text{if } load_i \leq 20\% \end{cases} \tag{5.17}$$

$$p_{\text{line}}(t) = \sum_{i=1}^{n_{\text{lines}}} p_{\text{line},i}(loading_i)$$

With $load_i$ is defined according to Chapter 2 and corresponds to the load of line i related to the maximum power flow capacity of the lines. $p_{\text{line},i}$ represents the penalty incurred by the line i. The utilization of each line is evaluated against staggered values to assign a penalty. The total line load penalty, p_{line}, is the sum of the penalties of all lines in the network. This approach incentivizes better line utilization and penalizes overload to prevent congestion and potential outages.

5.2.4 Balancing of Reward-Signal for Distinct Optimization Goals

From the preceding section, it is clear that the agent's objective is to devise an optimal strategy for scheduling ESS and CS and their applied power in order to minimize penalites, resp. to increase rewards. As detailed in the Section 5.2.3, this involves a spectrum of distinct penalties and rewards. Notably, all the penalties and reward signals discussed are positive scalars. Consequently, the signs of the individual penalties has to be reversed, to transform the penalty minimization problem into a reward maximization problem. With the sole exception of the reward from selling energy to the grid, which has naturally a positive impact on the overall reward signal. The ultimate goal is to minimize the penalties and thereby increase the total reward towards zero.

The overall reward signal $r(S_t, A_t)$ – obiously depending on states S_t and actions A_t – of the current time step t, composes as follows:

$$\begin{aligned} r(S_t, A_t)(t) = w_{\text{grid,r}} \cdot r_{\text{grid}}(t) - w_{\text{grid,p}} \cdot p_{\text{grid}}(t) \\ - w_{\text{cs}} \cdot p_{\text{cs}}(t) - w_{\text{pv}} \cdot p_{\text{pv}}(t) - w_{\text{line}} \cdot p_{\text{line}}(t) \end{aligned} \tag{5.18}$$

Whereas:

- $r_{\text{grid}}(t)$, reward for selling energy from grid, scaled by $w_{\text{grid,r}}$.
- $p_{\text{grid}}(t)$,penalty for buying energy from grid, scaled by $w_{\text{grid,p}}$.
- $p_{\text{cs}}(t)$, penalty for SoC on departure, scaled by w_{cs}.
- $p_{\text{pv}}(t)$, penalty for wasted PV-production, scaled by w_{pv}.
- p_{line}, penalty for exceeding line-load, scaled by w_{line}

The total reward signal in Equation (5.18) integrates all partial rewards and penalties, each modulated by an appropriate weight to determine its impact on the total reward and to guide the agent toward desired optimized power management strategies.

5.3 Environment Overview

This section provides an overview of the implemented RL environment based on the *gymasium* framework. Building on the agent-environment interface described in Section 2.1.2, the gymnasium library provides a framework for structuring a custom reinforcement learning problem. This allows a standardized interaction with external agents, e.g. to solve the problem with solvers from SB3.

The main functions defined within the environment can basically be divided into three stages. The Initialization, Reset and Step-function. First, the environment is initialized when the model is created. This includes setting the simulation parameters and preparing the environment for interaction with the agent. The second major function is the reset, at the beginning of each episode, which in our problem reflects a day, the necessary actions are taken to reset the environment to the beginning of the episode's day and to prepare it for the iterations over time steps for the next episode; this also happens at the very beginning of training, after the environment has been initialized. The third function is the step function. It basically represents the main interaction between the agent and the environment. At the end of each step, it informs the agent about the performance of its action as a reward/penalty signal and returns the observations for the next step, based on which the agent makes the next decision.

As described in Section 2.1.2, it is this interaction – the interplay of seeing the current state, deciding for an action based on it, receiving feedback in the form of a reward, which indicates how good the action taken was for the state observed – which allows the agent and its underlying, constantly optimized policy to make increasingly better decisions.

5.3.1 Initialization of the Environment

The initialization of the environment is the stage where the simulation parameters are set when the model is built from its environment class. It also prepares the environment for interaction with the agent. This subsection discusses the general process of initializing the environment, including setting up the state space, the action space, and all relevant parameters that define the dynamics of the environment.

The initialization is quite extensive, as it initializes and loads all the variables and data sets needed for training. For this reason, only the main points and peculiarities will be discussed. The initialization can be divided into three steps: Assigning variables and data, creating bus objects, and creating the observation & action space.

Variables and Data: Basically, variables are assigned here as part of the environment class *CsEnv*, which will be needed in many stages of the running simulation, allowing access to the model's values without having to constantly pass a multitude of variables to methods and functions.

Settings defined in the training configuration, such as the underlying scenario, i.e. the configuration of the distribution network with its buses and their components, whether pandapower network simulation is enabled – and if so, whether the previously created network json pandapower file matching the scenario is loaded –, whether it is an evaluation or a training run, and the weighting of the partial rewards, are assigned to the environment accordingly.

When training over many episodes, and especially when evaluating the results, it is important to have all details available. For this reason, internal simulation data

such as the current simulation date, the current time step, and the current training episode are also recorded.

In addition to the initialization data relevant for the scenario and the model, practical parameters such as the number of buses and the number of individual components are also identified and stored. This simplifies calculations in many places.

In addition, the entire prepared data set described in Section 3.2 for calculating the PV profiles and the residential load data set stored in *.csv* files are imported into the model during initialization. All data is accessible throughout a training run. At the beginning of each episode, the busses can directly access the data from the environment to determine the individual profiles, without the need for costly reloading from files at a later time.

Create Bus-Objects: Instances of the buses are created from the loaded scenario according to their configuration – see Section 4.2 for details. All bus objects created are stored in a single dictionary, which is useful as iteration is performed over the entirety of buses in many places during calculations in the step function, as well as during the reset and the associated retrieving of episode data.

Create Observation & Action Space: To provide a better overview and structure, certain functions of the *OptV2GEnv* class are outsourced to a helper class called *CsEnvHelper-Class*. Using Python's ability to merge one class into another, all methods of the helper class become part of the environment class.

This helper class contains methods needed to create the action- and observation-space, as well as to store the evolution of the simulation data for the current episode and record it in *.json* files for the purpose of evaluating the training process and the trained model. This includes all values needed to evaluate the time step of each episode, from the initialized state of the scenario – such as power values of individual loads and PV profiles, and initial states for ESS- & CS objects like (arrival-)SoC – to the decisions made by the agent and the resulting states of the environment. At the individual bus level, this includes power balances, the underlying consumption or feed-in values of the components, and at the global level, data such as the energy price over the episode, as well as the partial rewards and penalties.

5.3.2 Reset-Function

The name reveals the purpose of the function. Each episode, including the very first, starts with a reset of the environment. Each simulated episode is, at least in itself, considered and iterated separately. This is reasonable since, as mentioned above, an episode corresponds to a single real world day and the model is designed to optimize decisions by learning within a day.

In the last time step of an episode, i.e. also in the last iteration of the step function explained below, episode-specific variables such as the time step counter and the evolution histories of various variables within the previous episode have already been reset in order to provide a clean start for the next episode.

Then, when preparing a new episode, the data needed to represent a full day for the upcoming episode is loaded. This is done by randomly selecting a date from the range of available dates in the database of our environments. This date corresponds to the date simulated in the upcoming episode and is the basis for preparing the environment data and, accordingly, the data for the individual buses.

From the full set of day-ahead price data loaded when the environment is initialized, only the part of the selected episode date is retrieved. The same applies to the environment data for solar irradiance, temperature and cloud cover, which are used later to calculate the bus-specific PV profile.

For the household load profiles, the procedure is slightly different. Here, as described in section BSH, the database consists of only one year, for which data sets from 74 different households are available. In this case, the year of the episode date is ignored for the day to be simulated, and only the specific date is filtered from the entire dataset. Although this is a smaller historical period from which to choose, it still takes into account seasonal effects on load profiles, and the large number of 74 households results in a high degree of variability.

Once the episode data is loaded, it is iterated over all bus objects in the bus dictionary. For each existing bus, the bus-specific profile generation method is called, and depending on the availability of the individual PV and load components, a bus-specific load and PV profile is generated for the upcoming episode. For PV generation, this is done as described in the Section 3.2.2 section by incorporating

the ambient data of the episode. For the load profile, a random profile is selected from the entire episode data set consisting of 74 household load profiles for the same day. This results in highly varying profiles for different households across episodes, which does not necessarily reflect reality, but is intended to improve the generalization of the model during training.

The data for ESS- and CS-components are not based on real data sets, but are synthetically generated at the beginning of each episode and are based on probability distributions for individual time periods within an episode – see Section 3.2.4.

At the end of the reset, when all required episode data is available so that the environment is ready to provide the next episode, the environment method for obtaining observations, described in Section 5.2.1, is called. This retrieves the observations based on the episode data and allows the reset function to return them to the agent.

5.3.3 Step-Function

The Step-Function is called by the agent at each time step t of the currently simulated episode. The step begins by receiving the actions the agent decided based on the last observations. These are packed in a simple array, with a size depending on the defined and loaded scenario. As mentioned before, each bus can have a maximum of two controllable components, an ESS and a CS.

A key purpose of the Stepfunction is to apply the received actions to the corresponding components and thus calculating the new resulting state of the next time time step of the environenment. This is done in the outsourced function *simulate_actions*. The specific handling and application of the actions within the environment is described more detailed in the following Section 5.1.

As a follow-up task, the step function is also responsible for calculating the reward. In other words, it evaluates the actions applied to the current state of the network based on various criteria from Section 5.2.3 The calculated reward is composed of different partial rewards and penalties, which are weighted according to the optimization goal.

As it is not possible to evaluate a composed individual reward in order to understand the underlying partial rewards and penalties. All applied individual parts of the total reward are logged in a dictionary by the *getInfo*-function using the implemented custom callback mentioned in Section 4.2.2.

After applying the actions and maintaining the composite reward as a single scalar, the environment method is called at the end of each time step to provide the observations, as described in the *Reset-Function*. It provides the newly obtained environment state.

The observations for the state at the beginning of the next time step are returned together with the reward to the agent, which thus receives feedback on how good the decided actions made for the state observed actually were.

6 Testing and Validation of Approach

This section delves into testing and validating the developed model, which is pivotal for understanding its behavior, especially when assessing different weights within the reward function. This is instrumental in understanding how these adjustments influence the model's goals, performance and decision-making capabilities.

In order to be able to compare the results of different optimizations under variable conditions, a general scheme is first defined in Section 6.1 that allows the results of the model to be compared with different initial situations.

6.1 Scope and metrics for evaluation

To make the subsequent optimization sessions in Section 6.2 comparable, and also to be able to evaluate the scenarios from Section 7.1, the evaluation setting is introduced here. First, environmental conditions and a time horizon are defined that apply to all models to be evaluated, and then metrics are introduced to compare the trained models even under different optimization goals.

6.1.1 Evaluation Setting

The model is exposed to mostly random and non-reproducible conditions during training. To ensure comparability during evaluation, it is therefore important to create a consistent and reproducible environment. When evaluating a model that has been exposed to seasonally dependent, different days in the form of daily representative episodes, it makes sense to consider an entire year in the evaluation. This is also useful for general evaluation metrics of energy systems, since the billing of energy, be it PV feed-in or energy purchase, usually occurs in annual cycles. For

this purpose, the year 2019 of the dataset was chosen. Consequently, an evaluation session is carried out on the basis of 365 episodes over the entire year 2019. This includes the ambient data of solar irradiance, temperature and cloud cover as well as the basis of the load profiles. Any random noise in the creation of the profiles, in the creation of the initial data for CS and ESS, and in the selection of the load profiles was eliminated by the use of pseudo-random number generators. This ensures that for the same configuration of a scenario, each evaluation session will have exactly the same initial conditions.

6.1.2 Evaluation Metrics

To make the trained models comparable, the metrics used are briefly described hereafter. The metrics are based on reference values that would be used in a real context considering the domain of distribution grids, such as energy price per kWh, PV power in kW over a year. In addition, metrics are used that are based on the goals and rewards of the model, such as average SOC at exit, etc.

Average SoC on Departure

$$\begin{aligned} \bar{SoC}_i &= \frac{1}{n}\sum_{j=1}^{n} SoC_{i,j} \qquad \text{where } n = \text{departures at CS } i, \\ \bar{SoC}_{\text{overall}} &= \frac{1}{m}\sum_{i=1}^{m} \bar{SoC}_i \qquad \text{with } m = \text{total number of CS} \end{aligned} \tag{6.1}$$

The average (SoC) on departure for each car, denoted as $\bar{SoC}_i$, is calculated by taking the mean of all SoC values recorded at all departure times for specific car within the whole evaluation set, where n represents the total number of departures for car i. This average provides insight over the average SoC at departure of EVs at each CS.

The overall average SoC for the scenario, represented as $\bar{SoC}_{\text{overall}}$, is calculated by averaging these individual car averages $\bar{SoC}_i$ across all m CS-Components in the network. This overall average serves as a summary metric, offering a holistic view of the SoC of all CS-Components at the time of departure.

Average Price per kWh Load

During the model evaluation of scenarios with inhomogeneous bus configurations, it was found that different bus configurations result in 'unfair' ratios, since buses equipped with ESS and CS have free energy in each episode, which they get from the initial SOC of the components. Therefore, unfortunately, the following evaluation metric was introduced too late for a fairer evaluation of the results.

$$P_{\text{grid-req},i}(t) = \max\left(0, P_{\text{load},i}(t) - \left(P_{\text{pv},i}(t) + P_{\text{ess},i}(t) + P_{\text{cs},i}(t)\right)\right) \tag{6.2}$$

- $P_{\text{grid-req},i}(t)$ is the power requirement from the grid for bus i at time t.
- $P_{\text{load},i}(t)$ is the power load for bus i at time t.
- $P_{\text{pv},i}(t)$ is the power produced by photovoltaic systems for bus i at time t, with a negative sign since it contributes to reducing the load.
- $P_{\text{ess},i}(t)$ is the power interaction from the energy storage system for bus i at time t, which can be positive (discharging, thus supplying power) or negative (charging, thus adding to the load).
- $P_{\text{cs},i}(t)$ is the power interaction from the charging station for bus i at time t, which typically adds to the load and is thus positive.
- The use of the max function ensures that if the local sources produce more power than the load requires (resulting in a negative value inside the max function), the grid requirement is set to 0, indicating that no additional power needs to be drawn from the grid for that timestep.

Average PV-feed-in & proportion to generation

$$\begin{aligned} R_{\text{bus},i} &= \sum_{t=1}^{T} \frac{1}{4} E_{\text{to_grid},t,i} \cdot \text{da}_{\text{t}} - 0.05 \\ R_{\text{total}} &= \sum_{i=1}^{n_{\text{bus}}} G_{\text{bus},i} \\ \bar{R}_{\text{bus}} &= \frac{G_{\text{total}}}{n_{\text{bus}}} \end{aligned} \tag{6.3}$$

The total feed-in revenue for each bus, denoted as $R_{\text{bus},i}$, is calculated by summing up the products of the energy fed into the grid $E_{\text{to_grid},t,i}$ at each time step t and the corresponding real day-ahead price ($\text{da}_\text{t} - 0.05$) for that time step, across all time steps T within the evaluation set. The offset applies to the DA-price as described in Section 3.2.1. The division by 4 accounts for the 15-minute resolution, as prices are denoted per kWh.

The overall total feed-in revenue, represented as R_{total}, is obtained by aggregating the total gains $R_{\text{bus},i}$ from all buses i across the network, where n_{bus} is the total number of buses.

Moreover, the average feed-in revenue per bus, denoted as $\bar{G}_{\text{bus}}$, is determined by averaging the total gains $G_{\text{bus},i}$ over all n_{bus} buses. This average feed-in revenue serves as an metric of the typical economic return from energy feed-in per bus.

The cost of grid energy for each charging station at each timestep is calculated by multiplying cs_{from_grid} by the day-ahead energy price for that timestep, then dividing by 4 to adjust for the 15-minute intervals. This cost is then aggregated over all timesteps to determine the total cost for each charging station. Similarly, the total energy consumed (both from the grid and free sources) is aggregated.

The average price per kWh for the energy drawn by the charging stations is then calculated by dividing the total cost by the total energy consumed. This metric allows us to compare the cost-effectiveness of different charging strategies and configurations across the network.

The key formulas used in this calculation are as follows:

1. Total energy consumed by each charging station (E_{total}) is the sum of grid energy and free energy over all timesteps:

$$E_{total} = \sum_{t=1}^{T} \left(\frac{cs_{from_grid,t}}{4} + \frac{cs_{total,t} - cs_{from_grid,t}}{4} \right) \tag{6.4}$$

2. Total cost associated with grid energy consumption for each charging station (C_{total}) over all timesteps:

$$C_{total} = \sum_{t=1}^{T} \left(\frac{cs_{from_grid,t}}{4} \times price_t \right) \tag{6.5}$$

3. Average price per kWh for the energy drawn by the charging stations ($Price_{avg}$):

$$Price_{avg} = \frac{C_{total}}{E_{total}} \tag{6.6}$$

where $cs_{from_grid,t}$ is the power drawn from the grid at time step t, $cs_{total,t}$ is the total power consumed by the charging station at timestep t, and $price_t$ is the day-ahead price of energy at timestep t.

To optimize the following scenarios, the DDPG algorithm from the SB3 library was used in its default configuration, as shown in Table 6.1.

Parameter	Value
Learning Rate	0.001
Gamma	0.99
Action Noise	None (default)

Tab. 6.1: DDPG-Configuration.

6.2 Scenarios with Isolated Rewards

To pave the way for a more elaborate analysis of realistic and comprehensive examples of the forthcoming Chapter 7, in Section 6.2 the evaluation is based on simplified network configurations.

The exploration is organized around specific testing scenarios that aim to validate the individual components of the reward function. These scenarios are intentionally simple, allowing for a clear dissection of the model's response to various conditions.

All the following scenario configurations involve four buses, uniformly equipped, with identical load- and PV-profile distributions. The analysis is segmented into distinct conditions to demonstrate the influence of the reward function's weights:

- The first analysis in Section 6.2.1 takes place in an environment with CS-components at every bus, none of them is V2G-capable, there is no ESS-componets, concentrating solely on load, PV-generation and the charging of EVs. The reward function only considers unused PV-generation and energy prices through penalty p_{pv} and p_{grid}. It is anticipated that, under these circumstances, the optimal strategy will entail charging the buses exclusively during periods of high solar irradiance and low energy prices.

- The second analysis in Section 6.2.2 mirrors the first in terms of setup but adds another penalty p_{cs} towards encouraging charging activities. This adjustment is anticipated over different weights for p_{cs} to foster charging behaviors that are not exclusively confined to periods of high PV output, though such times are likely to be preferred.

- The third analysis in Section 6.2.3 is also based on the same network configuration, but will introduce V2G-capabilities for all CS-components in the first step, and in the second also consider an ESS at each bus. Since the dynamic is entirely changed, penalties of p_{cs}, p_{grid} and the reward r_{grid} for feeding-in energy to the grid will be considered.

- Lastly the most comprehensive configuration from third analysis including ESS will be trained again in Section 6.2.4, but this time also including the pandapower simulation and the thus available line load penalty p_{line} to gain insights in its influence.

Through these simplified scenarios, the operational logic of the model is unraveled, providing invaluable insights into its adaptability and efficiency under varied conditions. This foundational understanding helps for refining the approach and setting the stage for the more complex evaluations that follow.

Penalties: PV and Grid-Demand

The network is configured as depicted in Table 6.2. It is equally equipped at every bus, has equally configured PV-Systems, load-profiles and CS-power – V2G is deactivated for all CS.

The first validation scenario focuses only on two penalties. Firstly penalizing not self-consumed PV-production by p_{pv} – weighted with $w_{\text{pv}} = 3$ and secondly the penalty for buying energy from grid p_{grid} – weighted with $w_{\text{grid,p}} = 1$.

Config	#	Value
Buses	4	1
Load	4	1
PV	4	15 kW
CS	4	11 kW, no v2g

Tab. 6.2: Network Configuration PV.

This results in reward signal:

$$\begin{aligned} r(S_t, A_t)(t) &= -w_{\text{grid,p}} \cdot p_{\text{grid}}(t) - w_{\text{pv}} \cdot p_{\text{pv}}(t) \\ &= -1 \cdot p_{\text{grid}}(t) - 3 \cdot p_{\text{pv}}(t) \end{aligned} \tag{6.7}$$

As discussed in Section 5.2.3 the agent will be penalized for decisions which 'waste' PV-production and also for buying energy from the grid. Therefore, the agent is steered to control the charging station of mainly use PV-power to charge its car – set, it is available.

The resulting power values and charging schedule is shown exemplary for the CS-component at day 180 of the evaluation set (2019-06-30) of Bus 1 in Figure 6.1.

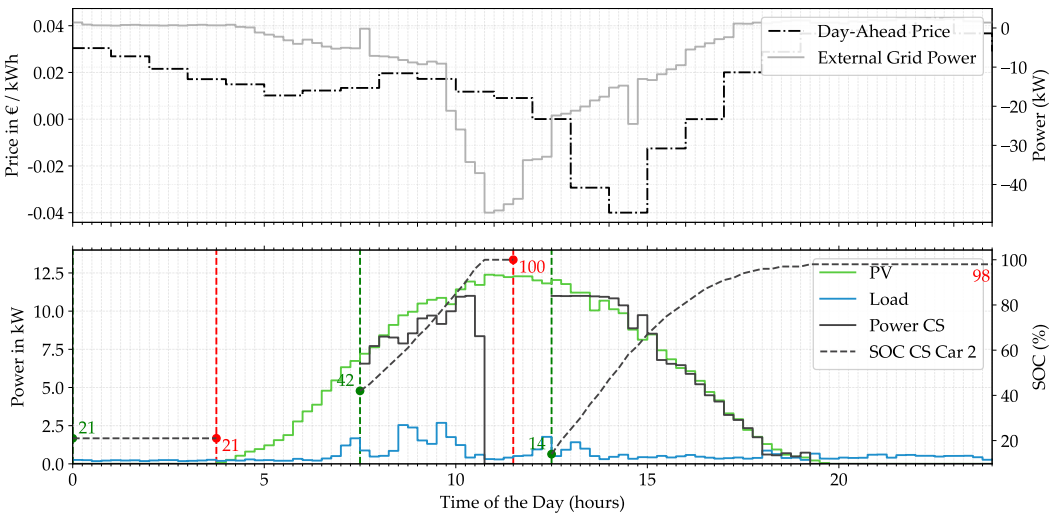
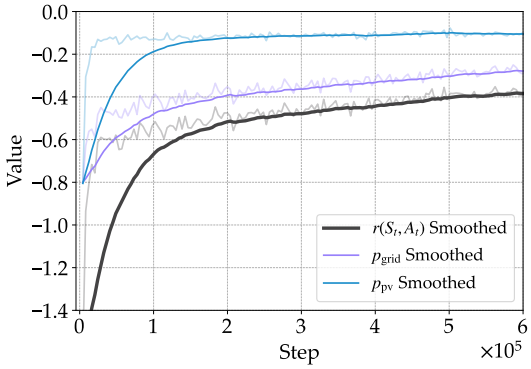


Fig. 6.1: CS-Schedule | Isolated Penalty p_{pv} and p_{grid} .

The lower plot in Figure 6.1 shows that the agent relies almost exclusively on the PV power available at its bus when deciding on the power at which the charging station operates in each time step, even as soon as the power of the load changes

and thus less PV power is available for charging, the CS power will adapt and try not to consume energy from the grid. The top plot shows a DA price drop that ends at hour 15. Given the relatively low price, the agent allows the CS to charge more than the available PV power during this period.



(a) Training Statistics for $r(S_t, A_t)$, p_{pv} and p_{grid} .

Metric	Value
Av. price price $\frac{€}{\text{kWh}}$	0.19
Av. price charging $\frac{€}{\text{kWh}}$	0.0072
Average SOC departure	51.70%
Wasted PV	46.22%

(b) Result Metrics.

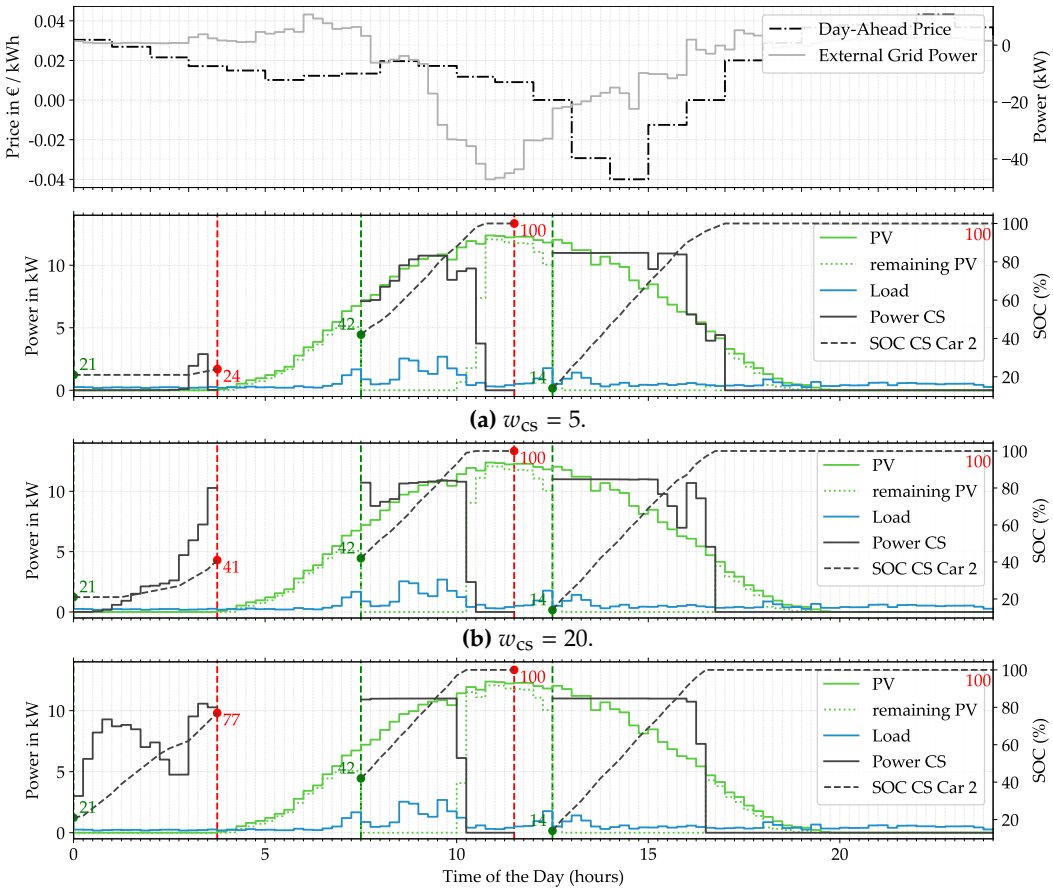
Fig. 6.2: Training statistics and Results | Penalties: PV and Grid-Demand.

Figure 6.2a shows the progression of the underlying partial rewards and the composite total reward over the course of the training. Their progression indicates that they do not contradict each other and improve in parallel over the course of the training. Table Figure 6.2b indicates how much a kWh costs to charge the vehicle, etc. These are averages calculated over the 365 evaluation episodes.

Penalties: PV, Grid-Demand and SoC

The same network configuration as in the first analysis from Table 6.2 is used again in the following. For training, the penalty for giving feedback of the SOC at the departure of the cars is added p_{cs} – weighted with $w_{\text{cs}} \in \{5, 20, 40\}$. This results in the reward signal:

$$\begin{aligned} r(S_t, A_t)(t) &= -w_{\text{grid,p}} \cdot p_{\text{grid}}(t) - w_{\text{cs}} \cdot p_{\text{cs}}(t) - w_{\text{pv}} \cdot p_{\text{pv}}(t), \\ &\quad \text{where } w_{\text{cs}} \in \{5, 20, 40\} \\ &= -1 \cdot p_{\text{grid}}(t) - w_{\text{cs}} \cdot p_{\text{cs}}(t) - 3 \cdot p_{\text{pv}}(t) \end{aligned} \tag{6.8}$$



(a) $w_{cs} = 5$.

(b) $w_{cs} = 20$.

(c) $w_{cs} = 40$.

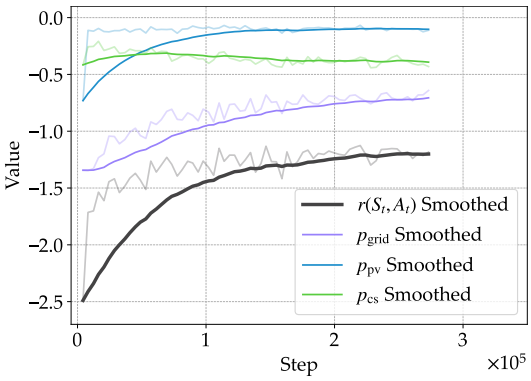
Fig. 6.3: CS-Schedule | Isolated Penalty p_{cs}, p_{pv} and p_{grid}.

Figure 6.3 clearly shows that for otherwise identical scenario conditions, the charging strategies differ only because of different values for w_{cs}, weighting the CS-penalty p_{cs}. Again, Bus 1 is representative for the visualization, speaking for all others.

In Figure 6.3a with $w_{cs} = 5$, the charging decisions still follow the PV production quite closely, only when energy prices fall, the vehicle is also charged again between hours 12-15. Even though the CS power follows the PV production very closely, remaining PV production is considered less, and the total consumption of load and CS now exceeds the PV production, as can easily be seen.

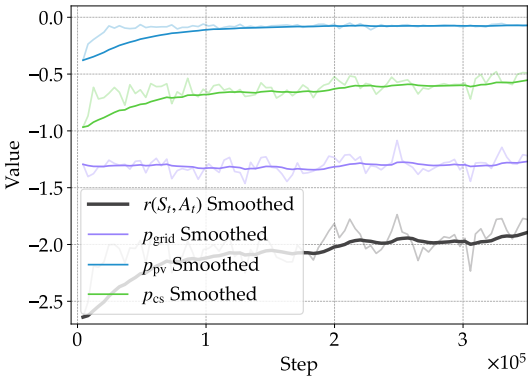
In Figure 6.3b with $w_{cs} = 20$ it can be observed that the falling DA-price is taken into account much more in the first charging sequence, resulting in reaching a much higher SoC of 41%.

In Figure 6.3c with $w_{cs} = 40$, the first charging sequence reaches the highest SoC of 71%. The energy price is no longer as important as the goal of achieving a high SoC. The behavior of the controller is not so easy to understand visually, so the training metrics and results for all three cases are shown in Figure Figure 6.4.



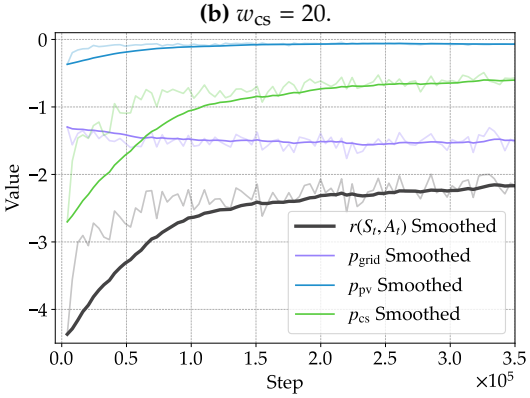
(a) $w_{cs} = 5$.

Metric	Value
Average SOC departure	72.83%
Wasted PV	52.08%
Average feed-in	353.60 €



(b) $w_{cs} = 20$.

Metric	Value
Average SOC departure	85.49%
Wasted PV	52.45%
Average feed-in	356.03 €



(c) $w_{cs} = 40$.

Metric	Value
Average SOC departure	90.52%
Wasted PV	53.99%
Average feed-in	367.42 €

Fig. 6.4: Training statistics and Results | Penalties: PV, Grid-Demand and SoC.

Figure 6.4 shows the progression of the underlying partial rewards and the composite total reward over the course of training for all three weight configurations. While p_{grid} first improves with the lower weights in 6.4a and remains stable in 6.4b,

it even decreases with a strong weight on p_{cs} in 6.4c. The grid-penalty gives the CS-penalty space over the course of training, allowing the EV to achieve a higher SoC on departure. p_{pv} still improves in every configuration. As expected, p_{cs} makes the most progress in improvement with the heavier weighting. This can also be seen in the increasing average SOC when leaving CS. From 72.83% to 85.49% to 90.52%. The progressions of p_{grid} and p_{cs} show that they are naturally contradictory, as the former tries to buy as little energy from the grid as possible, and the latter tries to achieve high SoC for all EVs connected to a CS.

V2G | Penalties: Grid-Demand and SoC | Reward: Grid

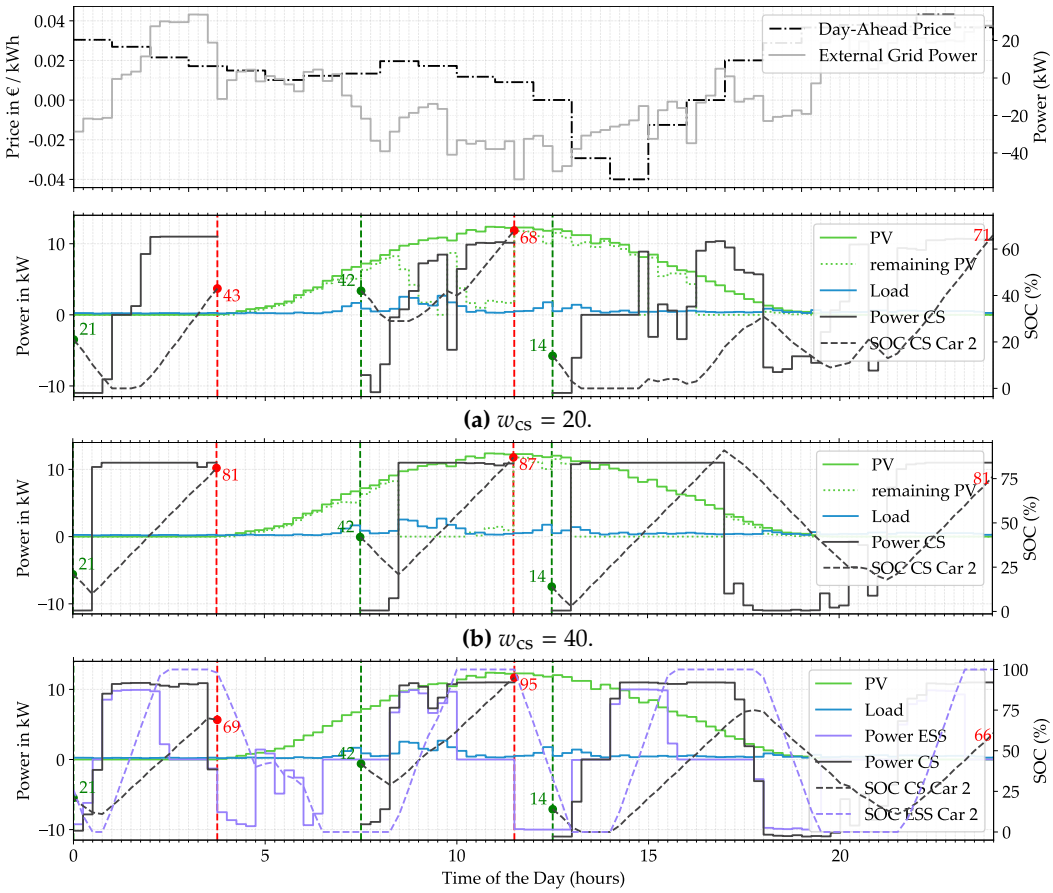
The next evaluation of the model follows essentially the same configuration as the last two test cases. The only difference is that V2G is enabled for all CS. This means that the CS are not only involved in consuming power, but can also supply power either to the components of the bus itself or to the surrounding grid. This creates a new dynamic, therefore in the following analysis the reward for selling energy r_{grid} is added, but the penalty for 'wasting' PV-production is dropped, resulting in this overall reward signal:

Config	#	Value
Buses	4	1
Load	4	1
PV	4	15 kW
CS	4	11 kW, v2g
ESS	4	10 kW, only in (c)

Tab. 6.3: Network Configuration V2G.

$$\begin{aligned} r(S_t, A_t)(t) &= w_{\text{grid,r}} \cdot r_{\text{grid}}(t) - w_{\text{grid,p}} \cdot p_{\text{grid}}(t) - w_{\text{cs}} \cdot p_{\text{cs}}(t), \\ &\quad \text{where } w_{\text{cs}} \in \{20,40\} \\ &= 1 \cdot r_{\text{grid}}(t) - 1 \cdot p_{\text{grid}}(t) - w_{\text{cs}} \cdot p_{\text{cs}}(t) \end{aligned} \tag{6.9}$$

Again, the exact same underlying scenario is used as before in Figure 6.5 to visualize the behavior of power and SoC at representative bus 1. In Figure 6.5a and 6.5b the different weighting of $w_{cs} = \{20,40\}$ is shown – this time considering V2G. In 6.5c, within the scenario of $w_{cs} = 40$, an ESS component has been added to each bus, resulting in another layer of observable power and SoC for the corresponding ESS-component.



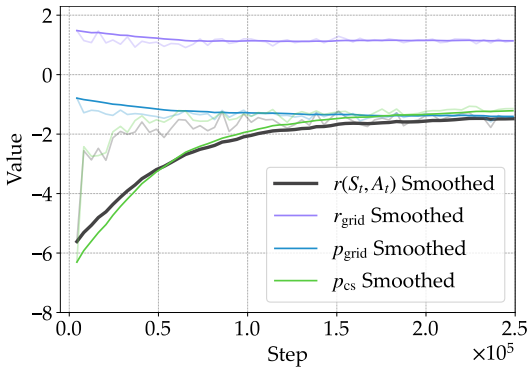
(a) $w_{cs} = 20$.

(b) $w_{cs} = 40$.

(c) including ESS | $w_{cs} = 40$.

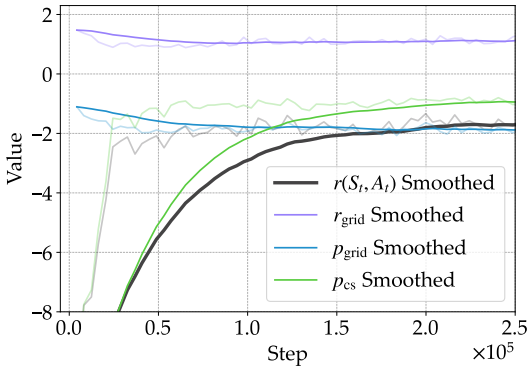
Fig. 6.5: V2G-Schedule | Isolated Penalty p_{cs}, p_{grid} and Reward r_{grid} | (c) considers ESS.

From Figure 6.5a and 6.5b we can see that adding V2G capability to the CS components brings a new dynamic. In periods of high energy prices, the performance becomes negative, indicating that the CS is feeding power into the grid, which of course happens more for the lower weight of $w_{cs} = 20$ in 6.5a, for $w_{cs} = 40$ in 6.5b, energy is still sold, but the focus is more on achieving high SoC. In 6.5c, the same is true, but since the ESS also wants to consume energy, the EV does not reach the same value of SoC in periods without PV production. The dynamics of the ESS are very similar to the power dynamics of the CS.



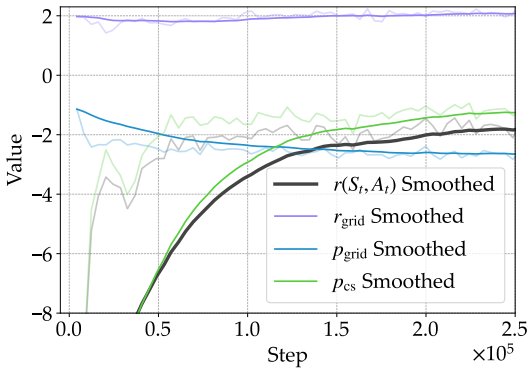
(a) $w_{cs} = 20$.

Metric	Value
Average SOC departure	73.075%
Wasted PV	57.60%
Average feed-in	3469.42 €



(b) $w_{cs} = 40$.

Metric	Value
Average SOC departure	85.49%
Wasted PV	52.45%
Average feed-in	3430.67 €



(c) including an ESS | $w_{cs} = 40$.

Metric	Value
Average SOC departure	84.79%
Wasted PV	50.09%
Average feed-in	5539.56 €

Fig. 6.6: V2G Training statistics and Results | Penalties: Grid-Demand, SoC and Reward: Grid | ESS included in (c).

Looking at the training statistics in Figure 6.6, it is shown that for all three configurations the positive reward r_{grid} and the negative penalty p_{grid} almost balance out at the end of training. For the lower weighting of w_{cs} in 6.6a, the total reward is

logically not quite as low as for the higher weighting of w_{cs} with and without ESS in 7.1b and 7.1c, respectively. The configuration in 7.1c that also includes ESS achieves both a lower value for the grid penalty and a higher value for the grid reward, which seem to balance each other out overall, as 7.1b and 7.1c approximately reach the same total reward in the last training step.

Penalties: Line, Grid-Demand and SoC | Reward: Grid

The last validation test includes a simulation of the power flow with *pandapower*. This evaluation run is therefore able taking into account the line loading of every connecting line between the buses in the network in every time step t. To do this, the same scenario from Section 6.2.3 (c)– including ESS-components at every bus – is trained again, this time considering the line penalty p_{line} from Section 5.2.3 with weight $w_{\text{line}} = 1$, resulting in the reward signal:

$$\begin{aligned} r(S_t, A_t)(t) &= w_{\text{grid,r}} \cdot r_{\text{grid}}(t) - w_{\text{grid,p}} \cdot p_{\text{grid}}(t) - w_{\text{cs}} \cdot p_{\text{cs}}(t) - w_{\text{line}} \cdot p_{\text{line}} \\ &= 1 \cdot r_{\text{grid}}(t) - 1 \cdot p_{\text{grid}}(t) - 20 \cdot p_{\text{cs}}(t) - 1 \cdot p_{\text{line}} \end{aligned} \tag{6.10}$$

For evaluation, the *pandapower* simulation is then applied to both trained models using the same configured scenario – but one has considered p_{line} during training, the other has not. This allows evaluating how training with p_{line} actually affects the line load.

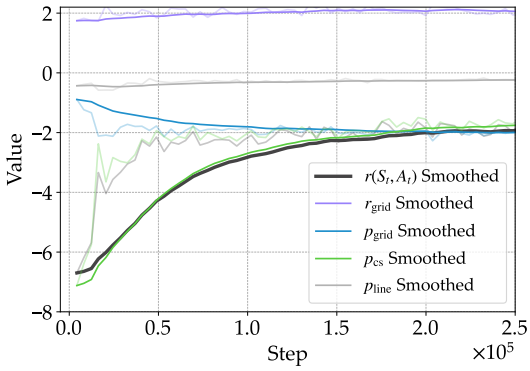
The interconnections through lines between the bus are fairly simple, represented by this matrix:

$$\text{Bus}_{\text{con}} = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{pmatrix} \tag{6.11}$$

Bus_{con} is a symmetric matrix used in the automatic generation of the consulted *pandapower* network as explained in Section 4.2.1 by the scenario configuration.

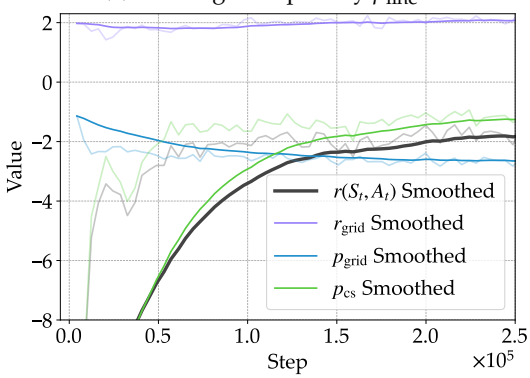
Each bus is connected to the bus before and after it, ending at the last bus number 4. This results in a radial distribution grid including three lines. [11]

Since none of the standard type lines in *pandapower* for 400V networks would reach their load limits for a small-scale distribution networks as in the scenario, a custom line was created for the evaluation in which the load limit $i_{\text{rated},i} = 0.05A$ for all lines i was set to half the value of the common standard type *15-AL1/3-ST1A 0.4* [24]. This is sufficient to occasionally exceed the 100% load limit of individual lines in the configured scenario and therefor serves the purpose of the evaluation.



(a) Training with penalty p_{line} .

Metric	Value
Average SOC departure	73.075%
Wasted PV	57.60%
Average feed-in	3469.42 €
Av. line loading	[38.2, 29.3, 21.8] %
Av. line loading	29.8 %



(b) Training without penalty p_{line}.

Metric	Value
Average SOC departure	85.49%
Wasted PV	52.45%
Average feed-in	3430.67 €
Av. line loading	[58.9, 42.3, 28.5] %
Av. line loading	43.2 %

Fig. 6.7: Line penalty p_{line} Training statistics and Results.

Figure 6.7 compares the training of the same scenario once with the line penalty in 6.7a and once without it in 6.7b. Looking at the progression of p_{line}, it is noticeable that it is very stable, at least in the scaling of the total reward $r(S_t, A_t)$. However, when comparing the two total reward progressions of the two training runs, it

becomes clear that this considerably changes the dynamics of the training. It also corrects p_{grid} upwards, which seems logical since applying a penalty for line loading implies reducing the overall consumption of the system. Looking at the metrics, it is clear that the penalty has the desired effect. On average, the line load is significantly reduced by 29.8 % for all three lines, compared to 43.8 % for the scenario without the penalty. Expressed in numbers in this perhaps rather arbitrary evaluation method, all three lines together were exceeded only in 16405 instead of 33346 of a total of 485450 15-minute time steps within the evaluated year. Looking at the average SoC at departure of 73 %, compared to 85 %, it is clear that p_{cs} has less influence and a restriction of the line loading is accompanied by compromises elsewhere.

6.3 Testing Evaluation Summary

Looking at the evaluated models of the different scenarios and according to logical understanding, it becomes clear that when modeling under consideration of ESS and CS components with activated V2G capability, the penalty for unconsumed PV generation is actually contradictory and only complicates the optimization. It seems to make little sense to prevent PV-power from being fed into the grid if the economic benefit of doing so outweighs the benefit of self-consumption. For an economically efficient scheduling with acceptable charge levels at departure, the inclusion of the penalty, which forces the cars to reach a reasnoable SoC p_{cs}, and the consideration of the energy price by r_{grid} and p_{grid} is sufficient.

7 Demonstration Developed Approach

The influence of the weighting of the different optimization objectives and the validity of the selected partial rewards and penalties have been studied using simple examples to better understand their dynamics. In the following, the developed approach is applied to more inhomogeneous scenarios, described in Section 7.1, with different bus configurations in more realistic environments in Section 7.1.1, and the results are evaluated in Section 7.1.2.

7.1 Scenario Selection

The network configurations chosen for the evaluation scenarios are based on a case study in [13]. The study outlines four scenarios for household technology adoption: Status Quo and three forward-looking scenarios for 2035 called Low, Medium, and High. These scenarios correspond to *German Network Development Plan* (NEP) scenarios A, B, and C for 2035, respectively. The Status Quo scenario is based on current data for rooftop PV installations, ESS, EVs. Deviating from the original reference, the presence of heat pumps is neglected in the following for simplicity. Households with and without heat pumps from the original case study are grouped accordingly. The future scenarios project the number of PV systems by maintaining the current average system size and the ratio of rooftop to ground-mounted systems. It is estimated that 90% of ESS will be installed in single-family and multi-family homes, and 75% of EVs will be residential. [13] Table 7.1 summarizes the distribution of the different technologies within the scenarios.

The scenarios predict a significant increase in EV penetration by 2035, with varying degrees of growth across the scenarios, while PV and BSS show less variation. The penetration rates, which reflect the proportion of households with specific

Configuration	Status Quo	Low 2035	Medi 2035	High 2035	Utopia 2035
	134 (91.8%)	92 (63.0%)	75 (51.4%)	48 (32.9%)	0 (0%)
	9 (6.2%)	8 (5.4%)	2 (1.4%)	0 (0%)	0 (0%)
	1 (0.7%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)
	0 (0%)	24 (16.5%)	40 (27.4%)	64 (43.8%)	0 (0%)
	1 (0.7%)	9 (6.2%)	13 (8.9%)	16 (10.9%)	0 (0%)
	1 (0.7%)	13 (8.9%)	16 (11%)	18 (12.3%)	146 (100%)

content from Utopia 2035 added separaetly.

Tab. 7.1: Number and share of households with different technologies (- only household, - PV system, - ESS, - EV) for all four scenarios (deviations from 100% result from rounding errors).[13].

technologies, illustrate these trends. A utopian scenario for 2035 is added for further analysis. In here, each household receives all available technologies. This will allow us to assess later whether this has a positive impact.

7.1.1 Scenario Design and Case Studies

The scenarios of the NEP and their technology distribution have been translated in order to model them well on a small scale distribution network within the framework of the model. Although the modeling of the scenarios within a configuration of 10 buses leads to deviations in the percentage distribution of the technology equipment, they still represent in principle the different expansion stages and their evaluation allows to draw conclusions on the different stages of the NEP. In the following, the application of the model to *Low 2035*, *High 2035* and *Utopia 2035* from table Table 7.1 are briefly explained.

Based on the results of the validation part in Section 6.3, and in order to simplify the training and the evaluation of results of the scenarios, for all three scenarios only the equal three partial rewards and penalties are considered, r_{grid}, p_{grid}, and p_{cs}. Also derived from the results of validation they are weighted with $w_{\text{grid,r}} = 1$, $p_{\text{grid,r}} = 1$ and $p_{\text{cs}} = 40$.

This produces the following reward signal:

$$\begin{aligned} r(S_t, A_t)(t) &= w_{\text{grid,r}} \cdot r_{\text{grid}}(t) - w_{\text{grid,p}} \cdot p_{\text{grid}}(t) - w_{\text{cs}} \cdot p_{\text{cs}}(t) \\ &= 1 \cdot r_{\text{grid}}(t) - 1 \cdot p_{\text{grid}}(t) - 40 \cdot p_{\text{cs}}(t) \end{aligned} \tag{7.1}$$

NEP-Scenario: Low

The Scenario from NEP *Low 2035*, can be seen in its version translated to 10 buses in Table 7.2.

Bus	Load	PV (kWp)	CS (V2G, kW, kWh)	ESS (kW, kWh)
Bus 0	X	-	-	-
Bus 1	X	-	-	-
Bus 2	X	-	-	-
Bus 3	X	-	-	-
Bus 4	X	12	-	-
Bus 5	X	-	V2G, 11, 50	-
Bus 6	X	-	V2G, 11, 50	-
Bus 7	X	12	V2G, 11, 50	-
Bus 8	X	12	V2G, 11, 50	10, 15
Bus 9	X	12	V2G, 11, 50	10, 15

Tab. 7.2: NEP-Scenario: Low 2035 Translated to a 10-Bus Network.

Three Households are fully equipped, with load-, PV-, CS- and ESS-component, one with load-, PV- and CS-component, two have only load- and CS-components, and the remaining four are only equipped wit loads.

NEP-Scenario: High

The Scenario from NEP *High 2035*, is summarized in its version translated to 10 buses in Table 7.3.

Bus	Load	PV (kWp)	CS (V2G, kW, kWh)	ESS (kW, kWh)
Bus 0	X	-	-	-
Bus 1	X	-	-	-
Bus 2	X	-	-	-
Bus 3	X	-	V2G, 11, 50	-
Bus 4	X	-	V2G, 11, 50	-
Bus 5	X	-	V2G, 11, 50	-
Bus 6	X	12	V2G, 11, 50	-
Bus 7	X	12	V2G, 11, 50	10, 15
Bus 8	X	12	V2G, 11, 50	10, 15
Bus 9	X	12	V2G, 11, 50	10, 15

Tab. 7.3: NEP-Scenario: High 2035 Translated to a 10-Bus Network.

Three households are fully equipped, with load-, PV-, CS- and ESS-component, two with load-, PV- and CS-component, three have only load- and CS-components, and the remaining three are only equipped wit loads.

Scenario: Utopia

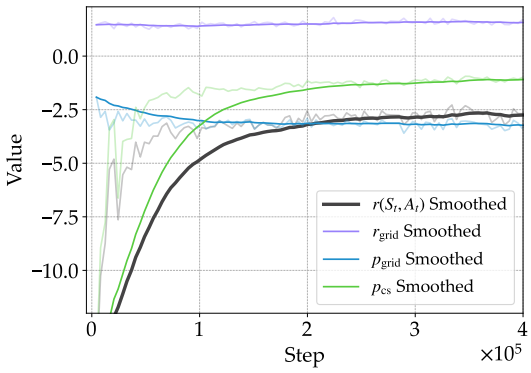
The Scenario *Utopia 2035*, is summarized in its version translated to 10 buses in Table 7.3.

Bus	Load	PV (kWp)	CS (V2G, kW, kWh)	ESS (kW, kWh)
Bus 0	X	12	V2G, 11, 50	10, 15
Bus 1	X	12	V2G, 11, 50	10, 15
Bus 2	X	12	V2G, 11, 50	10, 15
Bus 3	X	12	V2G, 11, 50	10, 15
Bus 4	X	12	V2G, 11, 50	10, 15
Bus 5	X	12	V2G, 11, 50	10, 15
Bus 6	X	12	V2G, 11, 50	10, 15
Bus 7	X	12	V2G, 11, 50	10, 15
Bus 8	X	12	V2G, 11, 50	10, 15
Bus 9	X	12	V2G, 11, 50	10, 15

Tab. 7.4: Scenario: Utopia 2035 Translated to a 10-Bus Network.

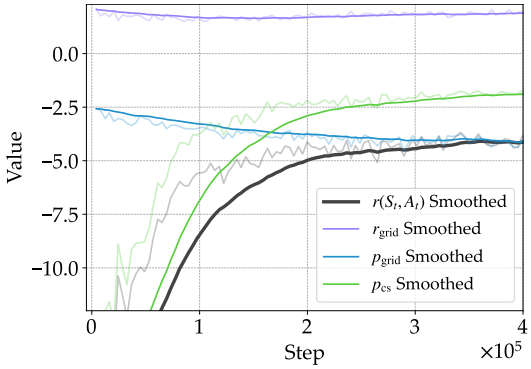
All households are fully equipped, with load-, PV-, CS- and ESS-component.

7.1.2 Evaluation of Scenario Results



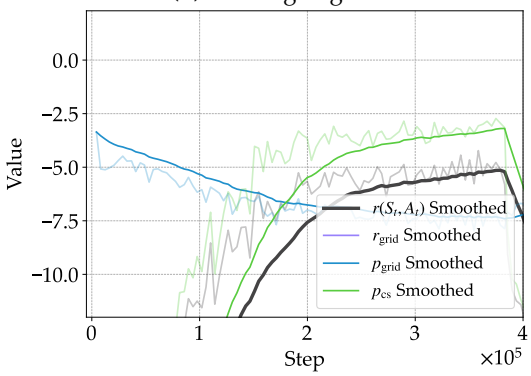
(a) $w_{\text{cs}} = 20$.

Metric	Value
Average SOC departure	86%
Wasted PV	54.79%
Average feed-in	2056.22 €



(b) Training High.

Metric	Value
Average SOC departure	86.41%
Wasted PV	43.09%
Average feed-in	2166.00 €



(c) Training Utopia.

Metric	Value
Average SOC departure	86.6%
Wasted PV	50.09%
Average feed-in	6423.73 €

Fig. 7.1: Training Stats Scenario Low 2035, High 2035 and Utopia 2035.

As can be seen in the training statistics in Figure 7.1 for the three scenarios, training is stable for all the different technology configurations of the future scenarios.

8 Discussion and Conclusion

In this work, the goal has been achieved to develop a comprehensive model to optimize and generate V2G-charging schedules for EVs in a network integrated approach. The underlying scenarios can be configured in a really flexible detail, in any network configuration with any configuration of buses, also the individual components of each bus can be configured according to the common configuration properties of their real counterpart. The underlying data used to model scenarios is a reasonable representation of reality and can also be easily replaced and modified, in general this is true for all components of the implementation. The approach is designed to capture the complexity of a real distribution network as closely as possible, using actual data for load and PV profiles in the simulation process. The power flow of the underlying network can also be simulated. This works well, but has not been sufficiently explored in this thesis. In terms of controlling the optimization, the design also allows for a lot of flexibility in selecting optimization targets and their characteristic influence to focus on, which also fulfills the original requirement. It has also been proven that the partial penalties work as intended. Along the way, however, it was discovered that perhaps not so many fine-grained partial rewards and penalties are needed, since the main requirement, at least in the context of these scenarios, is of course to achieve good economic results with high departure soc levels. This goes hand in hand with high PV self-consumption anyway, except at extremely low or high prices. The performance of the model was evaluated under different optimization strategies, but the scope of the work was unfortunately not sufficient to properly understand the optimization results of complex configurations in detail, although it is clear from the observed penalties that the results are quite likely to be in a valid range, at least considering the optimizations from the test scenarios. It was difficult to define metrics for the hard-to-understand dynamics of the power balances within a bus that would allow different-equipped buses to be evaluated against each other. This is a pity, as further

insight into the results would certainly be interesting. In addition, it was realized too late in the process that the evaluation runs, when trying to map a realistic year and define metrics that allow comparability, naturally become increasingly unfair as the buses get more equipped, since CS and ESS appear out of nowhere with a SOC that has nothing to do with the previous day or episode, providing additional free energy in addition to the PV supply, which makes comparability very difficult. While all of this is done in an effort to represent reality as closely as possible, it is important to note that certain assumptions made during the modeling process may not hold true in the real world. For example, the optimizations assumed an energy trading scenario that, while based on a realistic premise of buying and selling energy, involved a significant amount of arbitrage trading. This aspect of the model may not perfectly match the more complex realities of energy markets and EV usage patterns. In addition, the model did not fully account for factors that could significantly affect the dynamics of V2G systems in practice. These include the degradation of battery storage over time and the variability in the availability of connected EVs within the grid. Incorporating these factors into the model would likely change its performance and the resulting charging schedules, providing a different perspective on the optimization of V2G systems.

In conclusion, it must be said that a truly comprehensive approach has been developed that allows many different scenarios to be explored and, if well configured, mimics the real world to an appropriate degree. But it surely must be said that it is somewhat unfortunate that the full possibilities of evaluating complex scenarios hasent been exploited in the scope of this work. One important reason for this is certainly the lack of computational power, not being able to compute large and complex scenarios in a reasonable amount of time, the other certainly was prioritizing the application of the model too late by spending too much time in fine-tuning the configuration possibilities.

Despite these limitations, the developed model also serves as a valuable foundation for future research in this area. It highlights the potential of integrated V2G systems to contribute to the efficient management of energy resources within distribution networks, while also underscoring the importance of considering a wide range of variables to accurately model and optimize such systems.

8.1 Recommendations for Future Research

Without significant changes, the developed approach could be further explored and evaluated by exploiting its full potential in terms of scenario configuration. Much larger, more realistic systems with a large number of buses and different inhomogeneous component configurations are worth to be evaluated, as this has not been done sufficiently in the scope of this work. In addition, the already embedded power flow simulation may be used to gain much more detailed insight into the network conditions. Not many training configurations have been thoroughly tested yet. Different solvers and especially hyperparameter tuning could make the approach more efficient.

The implemented approach using partial rewards and penalties also offers a valid foundation for the implementation of multi-objective reinforcement learning approaches. In MORL, as in classical RL, there is only one agent making decisions, but the agent is exposed to separate reward signals that can be observed and responded to individually. Using a multi-agent strategy to optimize each bus individually according to its own requirements, rather than pursuing a single overarching goal, would also be desirable; this could be done in combination with a classical RL approach. Further , building on the existing and making the model more realistic by using more individual profiles, accounting for network losses, and eliminating more avoidable assumptions. It would also be interesting to use real-time market prices and real-time data in general, and as a result, a variant of the approach could be applied to a real network to make optimized loading decisions based on real states with real impact.

Appendix A

User Manual

Here is a brief explanation of how to use the environment.The repository is located at: . It contains a *requirements.txt* for installing the required packages. As soon as the packages are installed, the file *'solvers/train/universal_train_continue.py'* can be used to start and continue training.A configuration is already available by default. You can use *model_type* to choose from various algorithms of SB3, *session_prefix* allows you to freely select the name under which the model is logged and saved.*network_name* specifies the name of the *.json* file that contains a network configuration.These can be found in *'data/profile_json'*.Custom scenarios can be created according to their scheme *session_id* allows the specification of an id if more than one training session is carried out under the same prefix. in dict *objective_weights* the different weights of the penalties are configured for the training.The training can now be started.

Tensorboard must be used for a visual evaluation of the process.

For the evaluation and visualization, only *'solvers/evaluation/evaluate_trained_models.py'* must be executed with the respective session prefix and id.

Appendix B

Experimental Penalty

Experimental: Penalty for Overshooting (Dis-)Charging Decision

This Penalty is still part of the framework but won't be used for the underlying scenarios.

As described in energy calculations, the agent's action is clipped according to the maximum possible in view of physical restrictions, such as falling below or exceeding the valid SOC. Since the agent has knowledge of the current SOCs, a penalty is also introduced for this. If the agent's action exceeds the maximum possible charging and discharging capacity, it will be penalized accordingly.

As delineated in the energy calculations, the agent's actions are constrained by physical limitations, ensuring that the charging and discharging actions do not lead to states of charge (SoC) that fall below or exceed the permissible SoC range. To account for actions that attempt to exceed these physical constraints, a penalty mechanism is introduced. This penalty is imposed when the agent's proposed charging or discharging action exceeds the maximum feasible capacity determined by the current SoC and the charging station's capabilities.

The penalty for each CS is computed as the squared difference between the desired power ($p_{\text{desired},cs}$) based on the agent's action and the feasible power ($p_{\text{feasible},cs}$) that adheres to the physical constraints. For charging actions, $p_{\text{desired},cs}$ is calculated as the agent's action multiplied by the maximum power capacity of the charging

station ($cs_p_{\max,cs}$). The feasible charging power ($p_{\text{feasible},cs}$) is determined by the maximum charging power that does not lead to an SoC exceeding the maximum limit. Similarly, for discharging actions, $p_{\text{desired},cs}$ and $p_{\text{feasible},cs}$ are calculated considering the maximum discharging power that avoids reducing the SoC below zero. The penalty for each charging station is then given by:

$$\text{penalty}_{cs} = \begin{cases} (p_{\text{desired},cs} - p_{\text{feasible},cs})^2 & \text{if } p_{\text{desired},cs} > p_{\text{feasible},cs} \\ 0 & \text{otherwise} \end{cases} \tag{B.1}$$

This formulation ensures that penalties are only incurred for actions that attempt to exceed the charging or discharging capacities determined by the current state of the system. The total penalty for all charging stations is the sum of the individual penalties:

$$\text{Total Penalty} = \sum_{cs} \text{penalty}_{cs} \tag{B.2}$$

This penalty mechanism incentivizes the agent to make decisions within the feasible operational range of the charging stations and the vehicles' SoC levels, promoting efficient and realistic energy management strategies.

List of References

[1] Sozialdemokratische Partei Deutschlands and Bündnis 90/Die Grünen and Freie Demokratische Partei: *Koalitionsvertrag Zwischen SPD, Bündnis 90/Die Grünen Und FDP*. Online. 2021. URL: https://www.spd.de/fileadmin/Dokumente/Koalitionsvertrag/Koalitionsvertrag_2021-2025.pdf (cit. on p. 1).

[2] Author(s): *Mobility in Germany | Short Report Transport Volume – Structure – Trends*. BMVI, infas, DLR, IVT, infas 360. Bonn, Berlin, 2019. URL: https://bmdv.bund.de/SharedDocs/DE/Anlage/G/mid-2017-short-report.pdf?__blob=publicationFile (cit. on p. 1).

[3] ARBEITSGRUPPE 5: *Factsheet „vehicle to Grid" – Kundennutzen Und Netzintegration*. Nationale Plattform Zukunft der Mobilität, 2020. URL: https://www.plattform-zukunft-mobilitaet.de/wp-content/uploads/2020/10/201012_NPM_AG5_V2G_final.pdf (cit. on p. 1).

[4] Opinion of the European Economic and Social Committee on 'Communication from the Commission to the European Parliament, the Council, the European Economic and Social Committee and the Committee of the Regions — Powering a Climate-Neutral Economy: An EU Strategy for Energy System Integration' (COM(2020) 299 Final) () (cit. on p. 2).

[5] Bundesministerium für Wirtschaft und Energie: *Integrierter Nationaler Energie- Und Klimaplan*. Bundesministerium für Wirtschaft und Energie, 2020. URL: https://www.bmwk.de/Redaktion/DE/Downloads/I/integrierter-nationaler-energie-klimaplan.pdf?__blob=publicationFile&v=1 (cit. on p. 2).

[6] R. S. Sutton, A. G. Barto: *Reinforcement Learning: An Introduction*. Second edition. Adaptive Computation and Machine Learning Series. Cambridge, Massachusetts: The MIT Press, 2018. 526 pp. ISBN: 978-0-262-03924-6 (cit. on pp. 10–13).

[7] R. S. Sutton, A. G. Barto: *Reinforcement Learning: An Introduction*. Second edition. Adaptive Computation and Machine Learning Series. Cambridge, Massachusetts: The MIT Press, 2018. 526 pp. ISBN: 978-0-262-03924-6 (cit. on p. 13).

[8] D. Silver, G. Lever, N. Heess, T. Degris, D. Wierstra, M. Riedmiller: Deterministic Policy Gradient Algorithms () (cit. on p. 14).

[9] OpenAI: *Deep Deterministic Policy Gradient*. 2018. URL: https://spinningup.openai.com/en/latest/algorithms/ddpg.html (last access 03/03/2024) (cit. on p. 15).

[10] L. Thurner, A. Scheidler, F. Schafer, J.-H. Menke, J. Dollichon, F. Meier, S. Meinecke, M. Braun: Pandapower—An Open-Source Python Tool for Convenient Modeling, Analysis, and Optimization of Electric Power Systems. *IEEE Transactions on Power Systems* 33 (6) (2018), 6510–6521. ISSN: 0885-8950, 1558-0679. DOI: 10.1109/TPWRS.2018.2829021. URL: https://ieeexplore.ieee.org/document/8344496/ (last access 02/15/2024) (cit. on pp. 17–19, 40, 46).

[11] O. Ardakanian, V. W. S. Wong, R. Dobbe, S. H. Low, A. Von Meier, C. J. Tomlin, Y. Yuan: On Identification of Distribution Grids. *IEEE Transactions on Control of Network Systems* 6 (3) (2019), 950–960. ISSN: 2325-5870, 2372-2533. DOI: 10.1109/TCNS.2019.2891002. URL: https://ieeexplore.ieee.org/document/8601410/ (last access 03/16/2024) (cit. on pp. 19, 84).

[12] Ember Climate: *European Wholesale Electricity Price Data: Germany*. Ember Climate Data Catalogue. 2015–2023. URL: https://ember-climate.org/data-catalogue/european-wholesale-electricity-price-data/ (last access 02/29/2024) (cit. on p. 27).

[13] J. Stute, M. Kühnbach: Dynamic Pricing and the Flexible Consumer – Investigating Grid and Financial Implications: A Case Study for Germany. *Energy Strategy Reviews* 45 (2023), 100987. ISSN: 2211467X. DOI: 10.1016/j.esr.2022.1

00987. URL: https://linkinghub.elsevier.com/retrieve/pii/S2211467X2200181X (last access 02/28/2024) (cit. on pp. 27, 86, 87).

[14] *aWATTar Hourly Tariffs*. aWATTar. URL: https://www.awattar.de/tariffs/hourly (last access 02/29/2024) (cit. on p. 27).

[15] Solcast: *Global Solar Irradiance Data and PV System Power Output Data*. 2019. URL: https://solcast.com/ (cit. on p. 29).

[16] H. A. Kazem, J. H. Yousif: Comparison of Prediction Methods of Photovoltaic Power System Production Using a Measured Dataset. *Energy Conversion and Management* 148 (2017), 1070–1081. ISSN: 01968904. DOI: 10.1016/j.enconman.2017.06.058. URL: https://linkinghub.elsevier.com/retrieve/pii/S0196890417306027 (last access 02/24/2024) (cit. on p. 29).

[17] Deutsches Institut für Normung: *Photovoltaische Einrichtungen - Teil 3: Messgrundsätze Für Terrestrische Photovoltaische (PV)-Einrichtungen Mit Angaben Über Die Spektrale Strahlungsverteilung*. *Norm* DIN EN IEC 60904-3 VDE 0126-4-3:2020-01. Berlin, Germany: Deutsches Institut für Normung e.V., 2020 (cit. on p. 29).

[18] Solcast: *Solcast API Documentation*. 2024. URL: https://docs.solcast.com.au/ (cit. on p. 30).

[19] T. Tjaden, J. Bergner, J. Weniger, V. Quaschning: *Repräsentative Elektrische Lastprofile Für Wohngebäude in Deutschland Auf 1-Sekündiger Datenbasis*. 2015. DOI: 10.13140/RG.2.1.5112.0080/1 (cit. on p. 31).

[20] T. Tjaden, J. Bergner, J. Weniger, V. Quaschning: *Repräsentative Elektrische Lastprofile Für Einfamilienhäuser in Deutschland Auf 1-Sekündiger Datenbasis*. Hochschule für Technik und Wirtschaft (HTW) Berlin, 2015. URL: https://solar.htw-berlin.de/elektrische-lastprofile-fuer-wohngebaeude/ (last access 02/28/2024) (cit. on p. 31).

[21] M. Towers, J. K. Terry, A. Kwiatkowski, J. U. Balis, G. de Cola, T. Deleu, M. Goulão, A. Kallinteris, A. KG, M. Krimmel, R. Perez-Vicente, A. Pierré, S. Schulhoff, J. J. Tai, A. T. J. Shen, O. G. Younis: *Gymnasium*. Zenodo, 2023. DOI: 10.5281/zenodo.8127026. URL: https://zenodo.org/record/8127025 (last access 07/08/2023) (cit. on p. 39).

[22] A. Raffin, A. Hill, A. Gleave, A. Kanervisto, M. Ernestus, N. Dormann: Stable-Baselines3: Reliable Reinforcement Learning Implementations. *Journal of Machine Learning Research* 22 (268) (2021), 1–8. URL: http://jmlr.org/papers/v22/20-1364.html (cit. on p. 40).

[23] M. Abadi, A. Agarwal, P. Barham, E. Brevdo, Z. Chen, C. Citro, G. S. Corrado, A. Davis, J. Dean, M. Devin, S. Ghemawat, I. Goodfellow, A. Harp, G. Irving, M. Isard, Y. Jia, R. Jozefowicz, L. Kaiser, M. Kudlur, J. Levenberg, D. Mané, R. Monga, S. Moore, D. Murray, C. Olah, M. Schuster, J. Shlens, B. Steiner, I. Sutskever, K. Talwar, P. Tucker, V. Vanhoucke, V. Vasudevan, F. Viégas, O. Vinyals, P. Warden, M. Wattenberg, M. Wicke, Y. Yu, X. Zheng: *TensorFlow: Large-scale Machine Learning on Heterogeneous Systems*. 2015. URL: https://www.tensorflow.org/ (cit. on p. 40).

[24] pandapower: *Standard Types – Lines*. 2020. URL: https://pandapower.readthedocs.io/en/v2.3.0/std_types/basic.html#lines (cit. on p. 84).

Index

CPO, 2
CS, 2, 20, 23, 25, 32–34, 41, 45, 50, 55, 56, 59, 65, 74, 88–90, 96

DA, 50, 58, 72, 76, 78
DDPG, 14–16

EMSP, 2
EPEX, 27
ESS, 2, 20, 23, 25, 32, 33, 35, 41, 50, 55, 65, 74, 86, 88–90
EV, 1–5, 32–34, 59, 60, 86, 92

GTI, 29, 30

MDP, 9–12

NEP, 86, 87

PV, 3, 18, 20, 23, 29, 31, 41, 43, 50, 52, 73, 74

RL, 9, 10, 26, 39, 40, 42, 63

SB3, 14, 40, 63, 73, 95
SoC, 25, 33, 35, 45, 59, 65, 78, 80, 81, 85
STC, 29

V2G, 1, 2, 4, 5, 56, 74, 80, 81, 92
V2H, 2